

Decentralized Iris-Boosted Fuzzy Vault: Zero-Penalty Multimodal Template Protection with Threshold-Distributed Storage

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Abstract

Multimodal fuzzy vault schemes gate decoding on the auxiliary modality: when iris recognition fails, the system rejects the user even if the fingerprint alone would have succeeded. We propose the *Iris-Boosted Fuzzy Vault* (IBFV), which instead derives K bonus polynomial points from a stabilized iris key and injects them into the fingerprint vault. If iris recovery succeeds, the decoder gains K guaranteed-genuine points that shrink the Lagrange interpolation search space; if it fails, decoding falls back to the unimodal path with zero performance loss. We prove $\text{EER}_{\text{IBFV}} \leq \text{EER}_{\text{Uni}}$ and verify this across 960+ configurations on four FVC databases with CASIA-Iris-Interval, achieving up to 61.3% relative EER reduction. All four alternative architectures (AND-fusion, polynomial blinding, dense chaff, AES encryption) degrade performance; tightly coupled variants yield $\text{EER} \geq 19\%$.

We extend IBFV to a *Decentralized* variant (D-IBFV) that splits the vault into n shares via (t, n) -Shamir Secret Sharing over $\text{GF}(257)$ and distributes them across IPFS nodes with blockchain-anchored CID records. We prove that $t-1$ shares reveal zero information and t shares reconstruct the vault exactly. System experiments on Ganache (EVM) and Hyperledger Fabric 2.5 confirm enrollment under 806 ms, verification under 71 ms, and zero-bit-error reconstruction across 30,000 trials. The combined system satisfies all three ISO/IEC 24745 requirements with ECDLP-based irreversibility ($\sim 2^{128}$ operations).

Keywords: Biometric template protection, fuzzy vault, elliptic curve cryptography, multimodal biometrics, iris recognition, fingerprint, blockchain, Shamir secret sharing, decentralized identity

1. Introduction

Biometric authentication is increasingly deployed in border control, financial services, and national identity programs [1]. Unlike passwords, biometric traits are permanent: once a stored template is compromised, the underlying trait cannot be revoked. ISO/IEC 24745 [2] mandates three properties for biometric template protection (BTP): *irreversibility*, *unlinkability*, and *revocability*.

The fuzzy vault [3] is a widely studied BTP primitive that hides a secret polynomial among chaff points; recovery requires a sufficiently similar biometric query to identify enough genuine points for Lagrange interpolation [4]. Maurya *et al.* [10] ad-

vanced the scheme by mapping quantized minutiae to elliptic curve (EC) points over Brainpool P256r1 [11], achieving EER as low as 0.005% on FVC2002 DB1 with vault entropy exceeding 7.7 bits per coordinate. However, unimodal fingerprint systems remain vulnerable to spoof attacks and degrade on challenging data (EER 5.0% on FVC2002 DB3).

Multimodal fusion improves accuracy and spoof resistance [12], but integrating it into template protection is harder: the cryptographic binding must accommodate heterogeneous features while preserving security. Existing multimodal fuzzy vaults [9, 13] share a structural flaw: the second modality *gates* decoding. Under AND-fusion, $\text{GAR}_{\text{AND}} = \text{GAR}_{\text{FP}} \times \text{GAR}_{\text{Iris}} \leq \text{GAR}_{\text{FP}}$, so failure in the auxiliary modality rejects genuine users even when the fingerprint alone suffices. For iris subsystems with $\text{GAR}_{\text{Iris}} \approx 62\%$, over a third of

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genuine users are penalized.

We propose the *Iris-Boosted Fuzzy Vault* (IBFV), which uses the iris as an *optional booster* rather than a mandatory gate. A stabilized iris key seeds a PRF to derive K bonus polynomial evaluation points injected into the vault. If iris recovery succeeds, the decoder receives K guaranteed-genuine points that reduce the interpolation search space; if it fails, decoding proceeds identically to the unimodal system with zero penalty. We prove $\Pr[\text{Accept}_{\text{IBFV}}] \geq \Pr[\text{Accept}_{\text{Uni}}]$ for all configurations (Theorem 1).

Centralized vault storage remains a single point of failure. High-profile breaches—Aadhaar (1.1 billion records, 2018), the U.S. OPM (5.6 million fingerprints, 2015)—demonstrate the risk. We extend IBFV to a *Decentralized* variant (D-IBFV) that splits each vault via (t, n) -Shamir Secret Sharing [16] over GF(257), stores shares on IPFS [17], and records CIDs on a blockchain ledger.

Our contributions:

- A multimodal fuzzy vault with a provable zero-penalty guarantee—the first where the auxiliary modality never degrades performance—validated across 960+ configurations on four FVC databases with five architectures.
- Security analysis satisfying all three ISO/IEC 24745 requirements against Ratha *et al.*'s eight-point threat model [20], with ECDLP-based irreversibility ($\sim 2^{128}$ operations).
- A decentralized extension (D-IBFV) with threshold-distributed storage on IPFS and blockchain, with formal proofs of share confidentiality and exact reconstruction.
- Multi-platform system benchmarks on Ganache (EVM) and Hyperledger Fabric 2.5 confirming enrollment under 806 ms, verification under 71 ms, and zero-bit-error reconstruction across 30,000 trials.

Section 2 surveys related work. Section 3 reviews preliminaries. Section 4 presents IBFV. Section 5 describes D-IBFV. Section 6 provides security analysis. Sections 7 and 8 report biometric and system-level experiments. Section 9 concludes.

2. Related Work

2.1. Biometric Template Protection

BTP approaches fall into two categories. *Cancelable biometrics* [22] apply intentional, repeatable transformations to features before storage; Teoh *et al.* [23] introduced random multispace quantization for face and palm data. These methods provide revocability but may reduce discriminability. *Biometric cryptosystems* [24] bind a cryptographic key to biometric data. Dodis *et al.* [25] formalized this as *fuzzy extractors*; the fuzzy vault [3] and fuzzy commitment [8] are the most studied instantiations. Rathgeb and Uhl [4] survey both paradigms, noting that biometric cryptosystems offer stronger provable security but face greater challenges with noisy data.

2.2. Fuzzy Vault Schemes

The fuzzy vault [3] hides a degree- k secret polynomial among chaff points; decoding requires $\geq k+1$ genuine points for Lagrange interpolation. Clancy *et al.* [5] first applied it to fingerprint minutiae. Uludag *et al.* [7] added alignment helper data (EER 4.0% on FVC2002 DB2); Nandakumar *et al.* [6] reduced EER to $\sim 2\%$ via high-curvature alignment. Tams [26] analyzed record-multiplicity attacks, underscoring the need for enrollment-independent vault structures.

Maurya *et al.* [10] replaced traditional GF(2^{16}) encoding with ECC over Brainpool P256r1 [11], achieving EER as low as 0.005% (FVC2002 DB1) and vault entropy > 7.7 bits per coordinate. Our work builds on this ECC vault as the unimodal baseline.

2.3. Multimodal Biometric Fusion

Multimodal systems fuse information from multiple sources to improve accuracy and spoof resistance [12, 1]. Nandakumar and Jain [9] fused fingerprint and iris within a single fuzzy vault, but both modalities must be correct for decoding, introducing a gating penalty. Moon *et al.* [13] combined face and fingerprint similarly; the face module remains mandatory. Kelkboom *et al.* [27] fused multiple fingerprint algorithms within a fuzzy commitment framework but did not consider cross-modal fusion.

We additionally evaluate three tight-coupling strategies (Section 7.3): polynomial blinding, dense chaff seeding ($20\times$), and AES-256-GCM vault encryption. All three require the iris for decoding and exhibit the AND-fusion penalty.

2.4. Iris Recognition and Fuzzy Commitment

Daugman [28] established iris recognition via multi-scale Gabor encoding into binary iris codes, with fractional Hamming distance (FHD) as the matching metric. Hao *et al.* [8] combined iris codes with error-correcting codes via fuzzy commitment, binding a 140-bit key to a 2048-bit code with up to 25% error tolerance. We employ a repetition-code fuzzy commitment with bit interleaving over the full 131,072-bit iris code (Section 3.5), distributing spatially correlated errors across blocks for effective majority-vote decoding.

2.5. Blockchain-Based Biometric Systems

Delgado-Mohatar *et al.* [29] stored cancelable templates on-chain with high storage costs; Acquah *et al.* [14] identified off-chain storage with on-chain hash anchoring as the scalable pattern. Goel *et al.* [15] used Hyperledger Fabric for fingerprint templates (<1s latency) but stored unprotected data. Lee *et al.* [30] combined fuzzy extractors with Ethereum but required full templates on-chain. Sharma *et al.* [31] combined fingerprint/iris via cancelable transforms on blockchain (EER $\sim 8\%$), and Sharma and Dwivedi [32] deployed on Fabric without formal vault splitting. Yun *et al.* [33] applied zk-Rollup compression to template hashes, reducing gas by $\sim 90\%$ but without cryptographic vault protection. Prakasha *et al.* [34] identified three open challenges—vault-level splitting, formal threshold-security proofs, and cross-platform benchmarking—all addressed by D-IBFV.

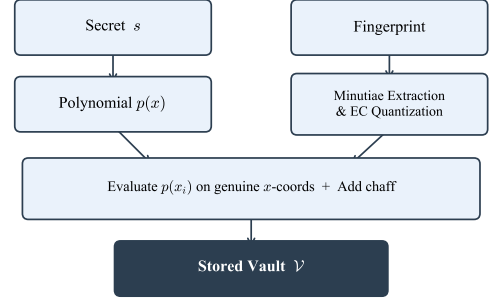
2.6. Summary and Gap Analysis

Table 1 summarizes the distinctions. All existing multimodal schemes require the auxiliary modality for decoding; failure implies rejection. IBFV is the first to decouple this trade-off: the iris provides a strict boost when available and zero penalty when unavailable. Existing blockchain-based schemes lack formal vault-splitting proofs and zero-penalty fusion. D-IBFV is the first to combine both.

3. Preliminaries

This section reviews the building blocks: elliptic curve cryptography (3.1), the fuzzy vault (3.2–3.3), iris extraction and commitment (3.4–3.5), Shamir’s Secret Sharing (3.6), and blockchain/IPFS infrastructure (3.7).

(a) Vault Locking



(b) Vault Unlocking

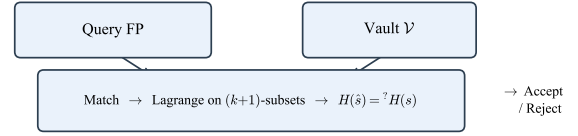


Figure 1: Fuzzy vault: (a) Locking—genuine points lie on $p(x)$, chaff are scattered. (b) Unlocking—query matches enable Lagrange interpolation.

3.1. Elliptic Curve Cryptography

An elliptic curve E over a prime field $\text{GF}(p)$ is defined by the short Weierstrass equation $y^2 = x^3 + ax + b \pmod{p}$. The points on E , together with the point at infinity $\mathcal{O}$, form an abelian group. Security rests on the *Elliptic Curve Discrete Logarithm Problem* (ECDLP): given G and $Q = s \cdot G$, finding s requires $O(\sqrt{n}) \approx 2^{128}$ operations for a 256-bit group. We use Brainpool P256r1 [11] ($p = 256$ bits, cofactor $h = 1$). The key property is that recovering a quantized minutia q from the vault point $P.x$ (where $P = q \cdot G$) requires solving ECDLP.

3.2. Fuzzy Vault Concept

The fuzzy vault [3] binds a secret to a noisy, unordered set (Fig. 1). A degree- k polynomial $p(x)$ is evaluated at biometric-derived coordinates; the resulting points are hidden among randomly generated chaff that do not lie on $p(x)$. Unlocking requires $\geq k+1$ genuine matches for Lagrange interpolation. Security relies on the indistinguishability of genuine and chaff points.

3.3. ECC-Based Fuzzy Vault

We adopt the ECC vault construction of [10]. Let E be the Brainpool P256r1 elliptic curve over $\text{GF}(p)$ with generator G and prime order n .

Table 1: Gap analysis: multimodal fuzzy vault and blockchain-based biometric architectures.

Scheme	Aux. Required	Gating Penalty	Zero-Penalty	Decentralized	Formal Splitting
Nandakumar [9]	Yes	Yes	No	No	—
Moon [13]	Yes	Yes	No	No	—
Kelkboom [27]	Yes	Yes	No	No	—
Arch. A (AND-fusion)	Yes	Yes	No	No	—
Arch. B (Blinding)	Yes	Yes	No	No	—
Arch. C (Dense Chaff)	Yes	Yes	No	No	—
Arch. D (AES Encrypt)	Yes	Yes	No	No	—
Delgado-Mohatar [29]	—	—	—	Yes	No
Sharma [31]	Yes	Yes	No	Yes	No
Lee (BDAS) [30]	—	—	—	Yes	No
Yun (Bio-Rollup) [33]	—	—	—	Yes	No
Sharma & Dwivedi [32]	—	—	—	Yes	No
IBFV (ours)	No	No	Yes	No	—
D-IBFV (ours)	No	No	Yes	Yes	Yes

3.3.1. Locking (Enrollment)

Given a set of fingerprint minutiae $\mathcal{M} = \{m_1, \dots, m_t\}$, the vault is constructed as follows:

1. **Quantization:** Each minutia $m_i = (x_i, y_i, \theta_i)$ is quantized by a factor f : $q_i = (\lfloor x_i/f \rfloor \bmod 64) \| (\lfloor y_i/f \rfloor \bmod 64) \| (\lfloor \theta_i/f \rfloor \bmod 64)$, producing an 18-bit integer. Duplicate quantized values are removed, yielding $\mathcal{Q} = \{q_1, \dots, q_g\}$ with $g \leq t$.
2. **EC Mapping:** Each q_i is mapped to an EC point $P_i = q_i \cdot G$, a one-way mapping protected by ECDLP.
3. **Polynomial Generation:** A random polynomial $p(x) = \sum_{j=0}^k a_j x^j$ of degree k is generated over $\text{GF}(p)$, with coefficients $a_j \in [10, 63]$ (6 bits each). The secret key is $s = \left(\bigoplus_{j=0}^k \text{bin}_6(a_j) \right) \bmod n$, and the public key is $S = s \cdot G$.
4. **Genuine Point Projection:** Each genuine minutia is projected onto the polynomial: $\mathcal{G} = \{(P_i.x, p(P_i.x))\}_{i=1}^g$.
5. **Chaff Generation:** $5g$ chaff points are generated: $\mathcal{C} = \{(c_j.x, r_j)\}_{j=1}^{5g}$ where $r_j \neq p(c_j.x)$, i.e., chaff points do *not* lie on $p(x)$.
6. **Vault Assembly:** All $6g$ points are randomly shuffled: $\mathcal{V} = \text{shuffle}(\mathcal{G} \cup \mathcal{C})$. The vault stores $(\mathcal{V}, S)$.

The security of the vault relies on the indistinguishability of genuine and chaff points: without

knowing which points lie on $p(x)$, an adversary must try $\binom{6g}{k+1}$ subsets.

3.3.2. Unlocking (Verification)

Given query minutiae $\mathcal{M}'$, quantize to $\mathcal{Q}'$ and find matching x-coordinates $\mathcal{X}_m = \{x : (x, y) \in \mathcal{V}, \exists q' \in \mathcal{Q}' \text{ s.t. } (q' \cdot G).x = x\}$. If $|\mathcal{X}_m| \geq k+1$, for each $(k+1)$ -subset of $\mathcal{X}_m$, perform Lagrange interpolation to reconstruct a candidate polynomial $\hat{p}(x)$, compute $\hat{s}$, and verify $\hat{s} \cdot G \stackrel{?}{=} S$. The vault is successfully unlocked if any subset yields a matching public key.

3.4. Iris Code Extraction

We implement a custom pipeline following Daugman [28] and Masek [35], operating on 320×280 NIR images from CASIA-Iris-Interval. **Segmentation:** The pupil is detected by intensity thresholding with contour refinement; the iris outer boundary is located via an integrodifferential-style radial gradient search. A noise mask excludes the pupil interior, specular reflections, and eyelid occlusions (detected via column-wise variance analysis on the normalized strip). **Normalization:** The annular iris region is unwrapped to a 64×512 strip using Daugman’s rubber-sheet model; the noise mask is simultaneously remapped. **Encoding:** The strip is CLAHE-enhanced and encoded with 1D Log-Gabor filters at two center wavelengths (18 and 36 pixels, $\sigma/f = 0.30$), producing a phase-quantized code of $64 \times 512 \times 2 \times 2 = 131,072$ bits. Fragile bits—near

phase boundaries where the response amplitude falls below $0.55 \times$ the median—are masked [36]. Mean validity coverage is $\approx 26\%$. **Matching:** Iris codes are compared via fractional Hamming distance (FHD) over mutually valid bits, with circular bit-shifts (± 32 columns) compensating for rotational misalignment.

3.5. Iris Fuzzy Commitment with Bit Interleaving

We extract a stable binary key from iris codes using a repetition-code fuzzy commitment. Let $\mathbf{c} \in \{0,1\}^N$ denote an N -bit iris code ($N = 131,072$ in our implementation) and let B_s denote the *block size* (number of bits per block). The number of blocks is $n_b = \lfloor N/B_s \rfloor$, each contributing one key bit, yielding an n_b -bit key.

3.5.1. Enrollment

1. Partition the code into n_b blocks using *interleaved* assignment: bit j belongs to block $(j \bmod n_b)$, distributing spatially correlated errors across blocks rather than concentrating them in contiguous regions.
2. Generate a random n_b -bit key $\mathbf{k} = (k_0, \dots, k_{n_b-1})$.
3. Construct a repetition codeword $\mathbf{w}$: position j is set to $k_{j \bmod n_b}$.
4. Compute helper data $\mathbf{h} = \mathbf{c} \oplus \mathbf{w}$ and store $(\mathbf{h}, H(\mathbf{k}))$ where H is SHA-256.

3.5.2. Recovery

Given query code $\mathbf{c}'$, for each rotation shift $\delta \in [-32, +32]$ (compensating for head tilt):

1. Compute $\tilde{\mathbf{c}} = \text{rotate}(\mathbf{c}', \delta)$ and $\tilde{\mathbf{w}} = \tilde{\mathbf{c}} \oplus \mathbf{h}$.
2. For each block i , recover $\hat{k}_i$ by majority vote over the block's valid positions (excluding masked bits). With $B_s = 255$, the error correction capacity per block is $(B_s - 1)/2 = 127$ bit errors (up to 49.8% error rate per block). Since genuine iris pairs exhibit $\text{FHD} \approx 0.20$ (i.e., ~ 51 errors per 255-bit block on average), majority vote succeeds for most blocks; however, all $n_b = 514$ blocks must decode correctly for key recovery, yielding an overall GAR of $\sim 62\%$.
3. If $H(\hat{\mathbf{k}}) = H(\mathbf{k})$, recovery succeeds; return $\hat{\mathbf{k}}$.

3.6. Shamir's Secret Sharing

Shamir's (t, n) -threshold Secret Sharing [16] distributes a secret $s \in \text{GF}(q)$ among n parties by evaluating a random degree- $(t-1)$ polynomial

$f(x)$ with $f(0) = s$ at n distinct points. Any t shares suffice for Lagrange reconstruction of s ; $t-1$ or fewer shares reveal zero information about s (information-theoretic security). We operate over $\text{GF}(257)$ —the smallest prime field containing all 256 byte values—applying Shamir sharing independently to each byte of the serialized vault.

3.7. Blockchain and IPFS Infrastructure

3.7.1. Hyperledger Fabric

Hyperledger Fabric [18] is a permissioned blockchain with pluggable consensus (Raft, BFT) and chaincode execution. Its execute-order-validate model achieves $>2,000$ TPS in optimized batch-transaction benchmarks [18]; per-invocation latency for complex chaincode (including IPFS I/O) is higher, as confirmed in Section 8.5. In D-IBFV, Fabric provides the tamper-evident CID ledger for permissioned deployments.

3.7.2. EVM-Compatible Platforms

Ethereum [19] and its Layer-2 derivatives (Polygon PoS [38], Arbitrum, Optimism) provide permissionless smart-contract execution. Any EVM-compatible chain can deploy the same Solidity contract. We use a local Ganache EVM instance as a controlled baseline.

3.7.3. IPFS

IPFS [17] is a content-addressed storage network where each object is identified by its SHA-256 hash (CID), providing integrity verification and decentralized availability. Each Shamir share is stored as an IPFS object; the CIDs are recorded on the blockchain.

4. Proposed Architecture: IBFV

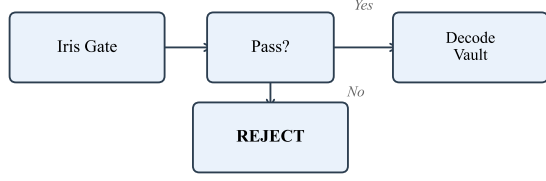
4.1. Motivation: The AND-Fusion Penalty

Consider a conventional multimodal fuzzy vault where the iris is used as a mandatory gate (e.g., Architecture A in our experiments). Let GAR_{FP} and GAR_{Iris} denote the genuine acceptance rates of the fingerprint and iris subsystems, respectively. Under AND-fusion:

$$\text{GAR}_{\text{AND}} = \text{GAR}_{\text{FP}} \times \text{GAR}_{\text{Iris}} \leq \text{GAR}_{\text{FP}} \quad (1)$$

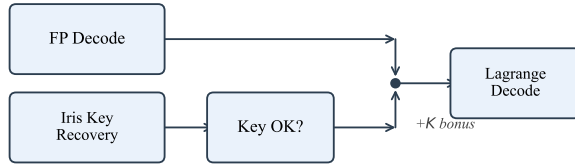
Since $\text{GAR}_{\text{Iris}} < 1$ in practice (our measurements show $\text{GAR}_{\text{Iris}} \approx 62\%$ for Setup 1), the AND-fusion GAR in (1) is always lower than the unimodal fingerprint GAR. Fig. 2 illustrates the difference between AND-fusion (gating) and IBFV (boosting).

(a) AND-Fusion (Gating)



$$GAR_{AND} = GAR_{FP} \times GAR_{Iris} \leq GAR_{FP}$$

(b) IBFV (Boosting)



$$GAR_{IBFV} \geq GAR_{FP} \quad (\text{iris can only help, never hurt})$$

Figure 2: AND-fusion vs. IBFV. (a) AND-fusion gates decoding on the iris; failure blocks the user. (b) IBFV uses the iris as an optional booster: success adds K bonus points; failure falls back to unimodal decoding with zero penalty.

4.2. IBFV Enrollment

Given fingerprint minutiae $\mathcal{M}$ and iris code $(\mathbf{c}, \mathbf{mask})$, the enrollment procedure (Algorithm 1, Fig. 3) proceeds as follows:

1. Execute the standard ECC vault enrollment (Section 3.3) to obtain genuine points $\mathcal{G}_{FP}$, polynomial $p(x)$, and public key S .
2. Perform iris fuzzy commitment enrollment (Section 3.5) with block size B_s to obtain commitment $(\mathbf{h}, H(\mathbf{k}))$ and key hash $\kappa = H(\mathbf{k})$.
3. **Derive virtual x-coordinates:** For $j = 0, \dots, K-1$:

$$v_j = \text{SHA-256}(\text{"ibfv_vpt:"} \| j \| \text{"} \| \kappa) \bmod p \quad (2)$$

ensuring $v_j \notin \{P_i.x : P_i \in \mathcal{G}_{FP}\}$ and $v_j \neq v_\ell$ for $j \neq \ell$.

4. **Compute bonus points:** $\mathcal{G}_{Iris} = \{(v_j, p(v_j))\}_{j=0}^{K-1}$.
5. **Generate chaff:** $5|\mathcal{G}_{FP}|$ chaff points (chaff count based on fingerprint points only).
6. **Construct vault:** $\mathcal{V} = \text{shuffle}(\mathcal{G}_{FP} \cup \mathcal{G}_{Iris} \cup \mathcal{C})$.

The stored vault is $(\mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s)$.

4.3. IBFV Verification

Given query fingerprint $\mathcal{M}'$ and iris code $(\mathbf{c}', \mathbf{mask}')$, verification (Algorithm 2, Fig. 4) proceeds:

IBFV Enrollment

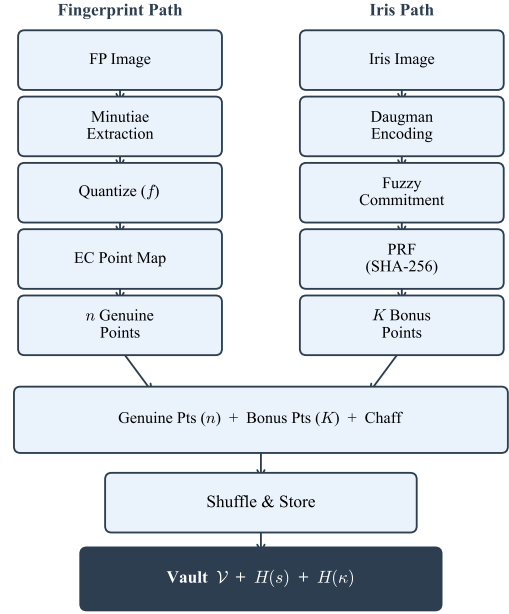


Figure 3: IBFV enrollment pipeline. The fingerprint path maps minutiae to EC points on Brainpool P256r1. The iris path extracts a key via fuzzy commitment, seeding a PRF to derive K bonus points. All points are merged with chaff.

1. Quantize query minutiae $\rightarrow$ find FP-matching vault points: $\mathcal{X}_{FP} = \{(x, y) \in \mathcal{V} : x \in \text{EC}(\mathcal{Q}')\}$.
2. Attempt iris key recovery (Section 3.5).
3. **If iris succeeds** (key $\hat{\mathbf{k}}$ recovered):
 - (a) Derive virtual x-coordinates $\{v_0, \dots, v_{K-1}\}$ using (2).
 - (b) Find bonus matches: $\mathcal{X}_{Iris} = \{(x, y) \in \mathcal{V} : x \in \{v_j\}\}$.
 - (c) Total candidate set: $\mathcal{X} = \mathcal{X}_{FP} \cup \mathcal{X}_{Iris}$.
 - (d) **Smart decoding:** Try $(k+1)$ -subsets of $\mathcal{X}$ that include *all* K bonus points first (since these are guaranteed genuine), reducing the search from $\binom{|\mathcal{X}|}{k+1}$ to $\binom{|\mathcal{X}_{FP}|}{k+1-K}$.
4. **If iris fails:** Set $\mathcal{X} = \mathcal{X}_{FP}$ and proceed with standard Lagrange interpolation—identical to the unimodal vault.

Complexity reduction. The smart decoding strategy exploits the fact that all K bonus points are *guaranteed genuine* (they were derived from the correct iris key). Instead of enumerating all $\binom{|\mathcal{X}|}{k+1}$ subsets, the decoder fixes the K bonus points and selects only the remaining $k+1-K$ points from

Algorithm 1 IBFV Enrollment

Require: Minutiae $\mathcal{M}$, iris code $(\mathbf{c}, \mathbf{mask})$,
 params (f, k, K, B_s)
Ensure: Vault $(\mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s)$

- 1: $\mathcal{Q} \leftarrow \text{QUANTIZE}(\mathcal{M}, f)$
- 2: $\{P_i\} \leftarrow \{q \cdot G : q \in \mathcal{Q}\}$ {EC mapping}
- 3: $p(x) \leftarrow$ random degree- k polynomial over $\text{GF}(p)$
- 4: $s \leftarrow \text{bin}(p) \bmod n$; $S \leftarrow s \cdot G$
- 5: $\mathcal{G}_{\text{FP}} \leftarrow \{(P_i.x, p(P_i.x))\}_{i=1}^g$
- 6: $(\mathbf{h}, \kappa) \leftarrow \text{IRISENROLL}(\mathbf{c}, \mathbf{mask}, B_s)$
- 7: **for** $j = 0$ **to** $K-1$ **do**
- 8: $v_j \leftarrow \text{SHA256}(\text{"ibfv_vpt:"} \| j \| \text{"} : \| \kappa \| \bmod p)$
- 9: **end for**
- 10: $\mathcal{G}_{\text{Iris}} \leftarrow \{(v_j, p(v_j))\}_{j=0}^{K-1}$
- 11: $\mathcal{C} \leftarrow \text{GENCHAFF}(5g)$ {5 $\times$ FP count}
- 12: $\mathcal{V} \leftarrow \text{SHUFFLE}(\mathcal{G}_{\text{FP}} \cup \mathcal{G}_{\text{Iris}} \cup \mathcal{C})$
- 13: **return** $(\mathcal{V}, S, \mathbf{h}, \kappa, B_s)$

the fingerprint candidate set $\mathcal{X}_{\text{FP}}$. This reduces the search space by a factor of:

$$\frac{\binom{|\mathcal{X}_{\text{FP}}|+K}{k+1}}{\binom{|\mathcal{X}_{\text{FP}}|}{k+1-K}} \quad (3)$$

For typical parameters ($|\mathcal{X}_{\text{FP}}| = 12$, $k = 10$, $K = 4$), this ratio is $\binom{16}{11} / \binom{12}{7} = 4368/792 = 5.5\times$, yielding a proportional speed-up in verification time while simultaneously *increasing* the probability of selecting a valid $(k+1)$ -subset.

4.4. Zero-Penalty Theorem

Theorem 1 (Zero-Penalty Guarantee). *For any parameter configuration (f, k) and any biometric sample pair $(\mathcal{M}', \mathbf{c}')$, the IBFV verification outcome satisfies:*

$$\Pr[\text{Accept}_{\text{IBFV}}] \geq \Pr[\text{Accept}_{\text{Uni}}] \quad (4)$$

where the probability is over the randomness in vault construction.

Proof. The IBFV verification process has two exclusive branches:

- **Iris succeeds:** The candidate set $|\mathcal{X}| = |\mathcal{X}_{\text{FP}}| + K \geq |\mathcal{X}_{\text{FP}}|$, and all K bonus points are on the polynomial. Since the unimodal decoder receives only $\mathcal{X}_{\text{FP}}$ while IBFV receives $\mathcal{X}_{\text{FP}} \cup \mathcal{X}_{\text{Iris}}$, every $(k+1)$ -subset available to the unimodal decoder is also available to IBFV, plus

Algorithm 2 IBFV Verification

Require: Query $(\mathcal{M}', \mathbf{c}', \mathbf{mask}')$, vault $(\mathcal{V}, S, \mathbf{h}, \kappa, B_s)$
Ensure: Accept / Reject

- 1: $\mathcal{Q}' \leftarrow \text{QUANTIZE}(\mathcal{M}', f)$
- 2: $\mathcal{X}_{\text{FP}} \leftarrow \{(x, y) \in \mathcal{V} : x \in \text{EC}(\mathcal{Q}')\}$
- 3: $\hat{\mathbf{k}} \leftarrow \text{IRISRECOVER}(\mathbf{c}', \mathbf{mask}', \mathbf{h}, \kappa, B_s)$
- 4: **if** $\hat{\mathbf{k}} \neq \perp$ **then**
- 5: **for** $j = 0$ **to** $K-1$ **do**
- 6: $v_j \leftarrow \text{SHA256}(\text{"ibfv_vpt:"} \| j \| \text{"} : \| H(\hat{\mathbf{k}}) \| \bmod p)$
- 7: **end for**
- 8: $\mathcal{X}_{\text{Iris}} \leftarrow \{(x, y) \in \mathcal{V} : x \in \{v_j\}\}$
- 9: $\mathcal{X} \leftarrow \mathcal{X}_{\text{FP}} \cup \mathcal{X}_{\text{Iris}}$
- 10: **else**
- 11: $\mathcal{X} \leftarrow \mathcal{X}_{\text{FP}}$ {Fallback: no penalty}
- 12: **end if**
- 13: **for each** $(k+1)$ -subset $\mathcal{S} \subseteq \mathcal{X}$ **do**
- 14: $\hat{p} \leftarrow \text{LAGRANGEINTERP}(\mathcal{S})$
- 15: $\hat{s} \leftarrow \text{bin}(\hat{p}) \bmod n$
- 16: **if** $\hat{s} \cdot G = S$ **then**
- 17: **return Accept**
- 18: **end if**
- 19: **end for**
- 20: **return Reject**

additional subsets containing bonus points. If the unimodal decoder accepts, so does IBFV.

- **Iris fails:** The fallback path uses $\mathcal{X} = \mathcal{X}_{\text{FP}}$ with identical Lagrange interpolation, producing the same outcome as the unimodal decoder.

In neither case does IBFV reject when the unimodal decoder would accept. $\square$ $\square$

Corollary 1. $\text{FRR}_{\text{IBFV}} \leq \text{FRR}_{\text{Uni}}$ for all (f, k) . *If the iris impostor acceptance rate is zero, bonus points are never injected for impostors, so $\text{FAR}_{\text{IBFV}} \leq \text{FAR}_{\text{Uni}}$ and therefore $\text{EER}_{\text{IBFV}} \leq \text{EER}_{\text{Uni}}$. In practice at $B_s = 255$, the iris impostor FAR is $\sim 0.10\%$ (Section 7.6), so a small fraction of impostors receive bonus points, potentially elevating FAR_{IBFV} above FAR_{Uni} by up to 0.07 pp on easy databases. The EER guarantee holds exactly when the iris impostor FAR is zero, and approximately when it is near-zero.*

IBFV Verification

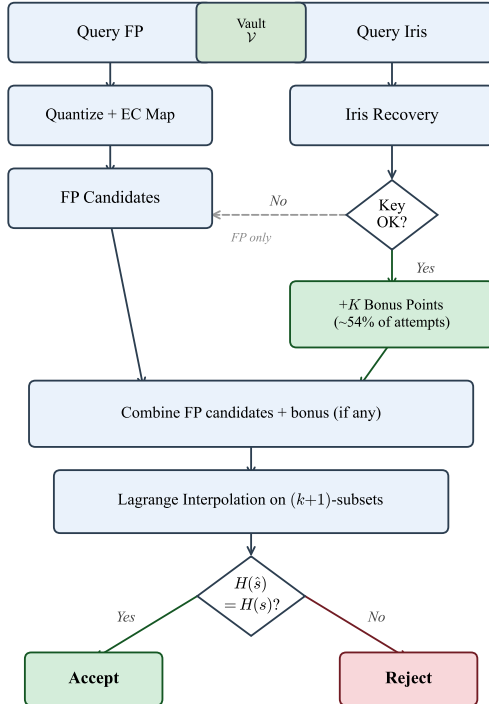


Figure 4: IBFV verification pipeline. If iris recovery succeeds (~62%), K bonus points reduce the search space. If it fails, the system falls back to unimodal decoding (dashed path).

5. Decentralized Storage Layer: D-IBFV

5.1. Motivation

The IBFV vault is a high-value target: a single breach exposes all users' protected templates. While the vault's cryptographic properties prevent biometric recovery, possession enables offline polynomial reconstruction attacks (Section 6.5). D-IBFV splits each vault into n shares via (t, n) -Shamir Secret Sharing and distributes them across independent nodes, targeting: (i) confidentiality ($t-1$ nodes reveal nothing), (ii) availability (t -of- n suffices, tolerating $n-t$ failures), and (iii) tamper-evidence via IPFS content addressing and blockchain CID anchoring.

5.2. Vault Serialization

The vault $(V, S, \mathbf{h}, H(\mathbf{k}), B_s)$ is deterministically serialized into a byte string $\mathbf{b} \in \{0, \dots, 255\}^L$ comprising: an 8-byte header; $|V| \times 64$ bytes of vault point coordinates; 64 bytes for S ; $\lceil N/8 \rceil$ bytes of iris helper data; 32 bytes for $H(\mathbf{k})$; 4 bytes of parameters; and a 32-byte SHA-256 integrity trailer.

Algorithm 3 D-IBFV Vault Splitting

Require: Serialized vault $\mathbf{b} \in \{0, \dots, 255\}^L$, threshold t , share count n

Ensure: Shares $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ where $\mathbf{s}_j \in \text{GF}(257)^L$

- 1: **for** $i = 0$ **to** $L-1$ **do**
- 2: Sample $a_1^{(i)}, \dots, a_{t-1}^{(i)} \xleftarrow{\$} \text{GF}(257)$
- 3: $f_i(x) \leftarrow b_i + a_1^{(i)}x + \dots + a_{t-1}^{(i)}x^{t-1} \pmod{257}$
- 4: **for** $j = 1$ **to** n **do**
- 5: $s_{j,i} \leftarrow f_i(j) \pmod{257}$
- 6: **end for**
- 7: **end for**
- 8: **return** $\{\mathbf{s}_j = (s_{j,0}, \dots, s_{j,L-1})\}_{j=1}^n$

For typical parameters ($g = 20$, $K = 4$, chaff = 5g), $L \approx 24$ KB.

5.3. Shamir Vault Splitting

Each byte b_i of the serialized vault $\mathbf{b} = (b_0, \dots, b_{L-1})$ is independently shared using (t, n) -Shamir Secret Sharing over $\text{GF}(257)$:

Theorem 2 (Share Confidentiality). *For any $t' < t$ compromised shares $\{\mathbf{s}_{j_1}, \dots, \mathbf{s}_{j_{t'}}\}$, the conditional distribution of the vault $\mathbf{b}$ is uniform over $\{0, \dots, 255\}^L$. That is, $H(\mathbf{b} \mid \mathbf{s}_{j_1}, \dots, \mathbf{s}_{j_{t'}}) = H(\mathbf{b})$.*

Proof. For each byte position i , the Shamir polynomial $f_i(x)$ has degree $t-1$ with $f_i(0) = b_i$. Given $t' < t$ evaluations $\{f_i(j_1), \dots, f_i(j_{t'})\}$, the polynomial has $t - t' \geq 1$ free coefficients, so $b_i = f_i(0)$ is uniformly distributed over $\text{GF}(257)$ [16]. Since the sharing is independent across byte positions, the joint distribution $(\mathbf{b} \mid \mathbf{s}_{j_1}, \dots, \mathbf{s}_{j_{t'}})$ is uniform. Because $b_i \in \{0, \dots, 255\} \subset \text{GF}(257)$, the adversary gains zero information about any byte. $\square$ $\square$

Theorem 3 (Exact Reconstruction). *Given any t shares $\{\mathbf{s}_{j_1}, \dots, \mathbf{s}_{j_t}\}$ with distinct indices, Lagrange interpolation recovers $\hat{b}_i = f_i(0) = b_i$ for all i , yielding $\hat{\mathbf{b}} = \mathbf{b}$ with zero bit errors.*

Proof. A polynomial of degree $t-1$ over $\text{GF}(257)$ is uniquely determined by t evaluations at distinct non-zero points. Lagrange interpolation at $x = 0$ yields $f_i(0) = b_i$ exactly. Since $b_i \in \{0, \dots, 255\} \subset \text{GF}(257)$, no truncation or modular artifact arises. $\square$ $\square$

5.4. Distributed Enrollment Protocol

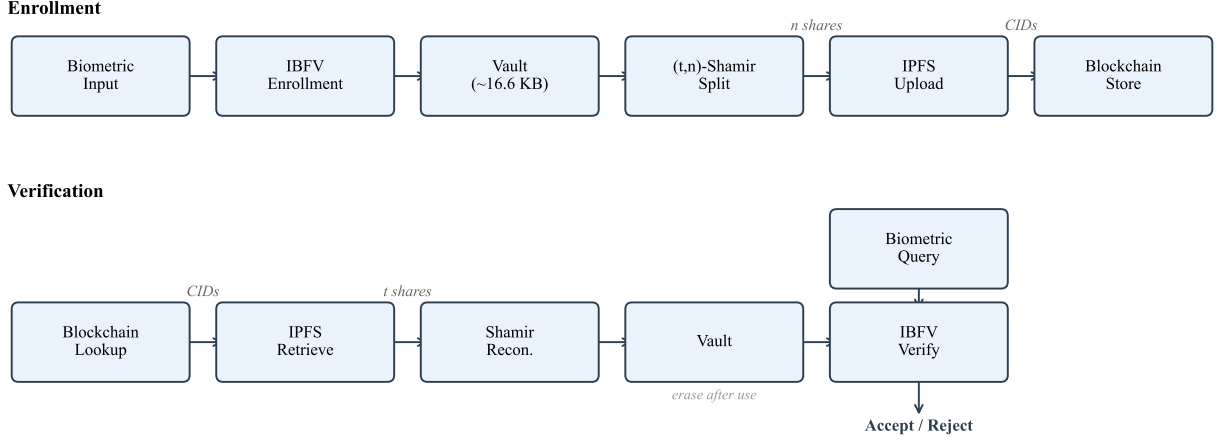


Figure 5: Complete D-IBFV architecture. **Top:** Enrollment pipeline—biometric data enters the IBFV enrollment, the vault is split via (t, n) -Shamir, shares are uploaded to IPFS, and CIDs are stored on the blockchain. **Bottom:** Verification pipeline—CIDs are retrieved from the blockchain, t shares fetched from IPFS, the vault reconstructed, and biometric verification performed. The reconstructed vault is erased after use.

Algorithm 4 D-IBFV Distributed Enrollment

Require: Biometric data $(\mathcal{M}, \mathbf{c}, \mathbf{mask})$, params (f, k, K, B_s, t, n)

Ensure: Blockchain transaction ID txid

- 1: $(\mathcal{V}, S, \mathbf{h}, \kappa, B_s) \leftarrow \text{IBFV-ENROLL}(\mathcal{M}, \mathbf{c}, \mathbf{mask}, f, k, K, B_s)$
 - 2: $\mathbf{b} \leftarrow \text{SERIALIZE}(\mathcal{V}, S, \mathbf{h}, \kappa, B_s)$
 - 3: $\{s_1, \dots, s_n\} \leftarrow \text{SHAMIRSPLIT}(\mathbf{b}, t, n)$ {Alg. 3}
 - 4: **for** $j = 1$ **to** n **do**
 - 5: $\text{cid}_j \leftarrow \text{IPFS-PUT}(s_j)$ {Upload share to IPFS}
 - 6: **end for**
 - 7: $\text{txid} \leftarrow \text{BLOCKCHAIN-STORE}(\text{uid}, \{\text{cid}_1, \dots, \text{cid}_n\}, t, n)$
 - 8: Securely erase $\mathbf{b}, \{s_j\}$ from local memory
 - 9: **return** txid
-

The enrollment protocol (Algorithm 4) extends IBFV enrollment (Algorithm 1) with three additional steps: serialization, Shamir splitting, and distributed storage. The complete vault never leaves the enrollment client in plaintext; only individual shares are uploaded to IPFS, and only their CIDs are recorded on the blockchain. After share distribution, the local copy of $\mathbf{b}$ is securely erased.

5.5. Distributed Verification Protocol

Verification (Algorithm 5) retrieves t shares from IPFS, reconstructs the vault via Lagrange interpolation, verifies the integrity checksum, and delegates biometric matching to the standard IBFV verification (Algorithm 2). The reconstructed vault exists in local memory only for the duration of the verification and is immediately erased afterward.

5.6. Revocation and Re-Enrollment

To revoke a compromised vault, the user:

1. Submits a revocation transaction to the blockchain, marking the current CID set as revoked (immutable audit trail).
2. Re-enrolls with fresh parameters (new polynomial, new iris commitment key, new Shamir randomness), producing a completely independent vault and share set.
3. Uploads new shares to IPFS and records new CIDs on the blockchain.
4. (Optional) Unpins old shares from IPFS nodes; even if old shares persist, the revocation record prevents their use for authentication.

The blockchain’s append-only nature ensures that revocation events are permanently recorded; an auditor can verify the complete enrollment/revocation history for any user. The fresh Shamir randomness in step 2 ensures that new shares are statistically independent of revoked shares (by Theorem 2).

Algorithm 5 D-IBFV Distributed Verification

Require: Query $(\mathcal{M}', \mathbf{c}', \mathbf{mask}')$, user ID uid**Ensure:** Accept / Reject

```
1:  $\{\text{cid}_1, \dots, \text{cid}_n\}, t \leftarrow \text{BLOCKCHAIN-LOOKUP}(\text{uid})$ 
2: Select any  $t$  available CIDs:  $\mathcal{T} \subseteq \{1, \dots, n\}$ ,  $|\mathcal{T}| = t$ 
3: for  $j \in \mathcal{T}$  do
4:    $\mathbf{s}_j \leftarrow \text{IPFS-GET}(\text{cid}_j)$  {Download share}
5:   Verify  $\text{CID}(\mathbf{s}_j) = \text{cid}_j$  {IPFS integrity check}
6: end for
7:  $\hat{\mathbf{b}} \leftarrow \text{SHAMIRRECONSTRUCT}(\{(j, \mathbf{s}_j)\}_{j \in \mathcal{T}}, t)$ 
8: Verify  $\text{SHA256}(\hat{\mathbf{b}}[0 : L-32]) = \hat{\mathbf{b}}[L-32 : L]$  {Integrity}
9:  $(\mathcal{V}, S, \mathbf{h}, \kappa, B_s) \leftarrow \text{DESERIALIZE}(\hat{\mathbf{b}})$ 
10: result  $\leftarrow \text{IBFV-VERIFY}(\mathcal{M}', \mathbf{c}', \mathbf{mask}', \mathcal{V}, S, \mathbf{h}, \kappa, B_s)$ 
11: Securely erase  $\hat{\mathbf{b}}, \{\mathbf{s}_j\}$  from local memory
12: return result
```

5.7. Extended Threat Model

D-IBFV extends the eight-point threat model (Section 6.1) with three additional attack points specific to the decentralized architecture:

9. **Node compromise (AP-9):** An adversary compromises up to $t-1$ storage nodes, obtaining their shares. *Countermeasure:* Theorem 2 guarantees zero information leakage from $t-1$ or fewer shares.
10. **Blockchain tampering (AP-10):** An adversary attempts to modify CID records on the blockchain. *Countermeasure:* The append-only, consensus-protected ledger makes retroactive modification computationally infeasible (51% attack for PoW/PoS; $f < n/3$ Byzantine nodes for BFT). IPFS content addressing provides independent integrity verification: any CID modification produces a hash mismatch upon share retrieval.
11. **IPFS share substitution (AP-11):** An adversary replaces a share blob on IPFS with a crafted payload. *Countermeasure:* IPFS CIDs are cryptographic hashes of the content; substituting the content changes the CID. The blockchain-anchored CID serves as a tamper-evident reference. Additionally, the vault's integrity checksum (SHA-256 trailer) detects any reconstruction from corrupted shares.

Table 2: Attack points following Ratha *et al.* [20].

AP	Attack	Countermeasure
1	Spoofing	Liveness detection (complementary)
2	Sensor interception	Client-side processing
3	Extractor compromise	Trusted computing base
4	Feature channel	Local processing, no transmission
5	Hill-climbing	Binary output only (no scores)
6	Database breach	§6.2–6.5
7	Template replay	Requires live biometric query
8	Decision override	Transport-layer security

6. Security Analysis

We analyze IBFV against ISO/IEC 24745 requirements using the eight-point adversary model of Ratha *et al.* [20].

6.1. Threat Model

Following Ratha *et al.* [20], we consider eight attack points in the biometric pipeline (see Fig. 6):

Table 2 summarizes the eight attack points. AP-1 through AP-4, AP-7, and AP-8 are mitigated by standard system-level measures (liveness detection, secure channels, trusted computing base). AP-5 is countered by the binary Accept/Reject output, which exposes no gradient signal for hill-climbing [20]. The primary threat is **AP-6 (database breach)**: the adversary obtains the stored template $(\mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s)$ in its entirety, addressed in Sections 6.2–6.5.

We assume the adversary has full access to the stored template (AP-6) and the iris helper data, but not the original biometric samples (standard stolen-token model). For D-IBFV, we additionally consider adversaries compromising up to $t-1$ storage nodes (AP-9), the blockchain (AP-10), or IPFS shares (AP-11), as defined in Section 5.7.

6.2. Irreversibility (Non-Invertibility)

We establish that three independent barriers prevent template inversion.

6.2.1. Fingerprint Recovery from Vault

Given vault $\mathcal{V}$ containing g genuine and $5g + K$ non-genuine points, the adversary must identify the genuine subset. Without knowledge of which points lie on the polynomial $p(x)$, the adversary faces a combinatorial identification problem. Even after identifying $k+1$ genuine points, recovering the original minutiae requires inverting the EC mapping: given $P_i.x$, finding q_i such that $(q_i \cdot G).x = P_i.x$ is an instance of ECDLP.

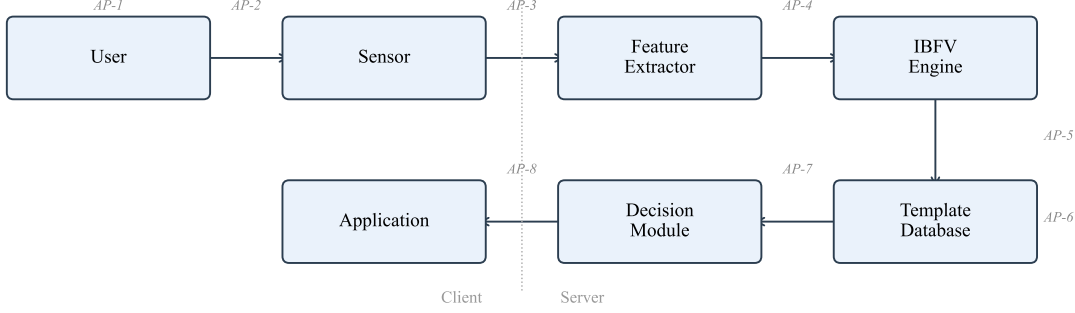


Figure 6: Threat model: eight attack points in the IBFV pipeline, following Ratha *et al.* [20]. AP-6 (database breach) is the focus of our analysis.

For Brainpool P256r1 with a 256-bit group order, the best known attack (Pollard’s rho method) requires approximately:

$$\sqrt{\pi n/2} \approx 2^{128} \text{ group operations} \quad (5)$$

At 10^9 operations per second, this requires approximately 10^{29} seconds ($\sim 3 \times 10^{21}$ years), providing a wide margin above the 2^{80} threshold typically considered infeasible.

6.2.2. Iris Code Recovery

The helper data $\mathbf{h} = \mathbf{c} \oplus \mathbf{w}$ reveals no information about the iris code $\mathbf{c}$ without knowing the codeword $\mathbf{w}$. The space of valid codewords is $\{0,1\}^{n_b}$ (repeated to length N), giving 2^{n_b} possibilities. For $B_s = 255$ (giving $n_b = \lfloor 131072/255 \rfloor = 514$ key bits), the brute-force cost is $2^{514} \approx 10^{155}$ hash evaluations—completely infeasible.

6.2.3. Bonus Point Identification

Without the iris key hash $\kappa = H(\mathbf{k})$, the adversary cannot compute the virtual x-coordinates (2) due to the preimage resistance of SHA-256. The K bonus points are computationally indistinguishable from chaff and genuine fingerprint points in $\text{GF}(p)$, as they are uniformly distributed 256-bit integers (by the PRF property of SHA-256). This adds K additional unknowns to the vault identification problem.

6.3. Unlinkability

Given two vaults $\mathcal{V}_1, \mathcal{V}_2$ enrolled from the same user in different applications, unlinkability requires that no polynomial-time adversary can determine

whether the vaults share a common biometric source.

IBFV achieves unlinkability through three independent mechanisms:

1. **Random polynomial:** Each enrollment uses an independently sampled polynomial $p(x)$, so the y -coordinates $(p(P_i.x))$ are statistically independent across enrollments.
2. **Random chaff:** Chaff points are uniformly generated in each enrollment, providing no cross-vault correlation.
3. **Iris-derived coordinates:** The virtual x-coordinates $\{v_j\}$ depend on $\kappa = H(\mathbf{k})$, which is deterministic per enrollment. However, if different applications use different quantization/commitment parameters (standard practice for revocability), the resulting κ values differ, producing independent PRF outputs.

While the *genuine x-coordinates* repeat across enrollments (since they derive from the same biometric), these are hidden among $5g + K$ non-genuine points. Identifying common genuine points across two vaults requires solving the intersection problem on shuffled sets—equivalent to the polynomial reconstruction attack, which is combinatorially hard (Section 6.5).

6.4. Revocability

If a vault is compromised, the user re-enrolls with a new random polynomial, new random seed for quantization (*e.g.*, different factor f or domain-specific salt), and a fresh iris commitment (using a different random key $\mathbf{k}'$). The new vault’s genuine

point projections $p'(P_i.x)$, chaff, and iris-derived bonus coordinates are all independent of the revoked vault. Multiple revocations are supported without biometric re-acquisition—only the computational parameters change.

6.5. Brute-Force Attack Analysis

We quantify the computational cost of a brute-force attack on a stolen vault.

6.5.1. Polynomial Reconstruction Attack

The adversary must identify at least $k+1$ genuine points among $|\mathcal{V}| = 6g + K$ total points. With typical parameters ($g \approx 20$, $K = 4$, $k = 10$), the vault contains $6(20) + 4 = 124$ points, of which $20 + 4 = 24$ are genuine. The number of candidate $(k+1)$ -subsets is:

$$\binom{124}{k+1} = \binom{124}{11} \approx 1.7 \times 10^{15} \quad (6)$$

Each candidate requires one Lagrange interpolation ($O(k^2)$ field operations) and one EC scalar multiplication ($\hat{s} \cdot G$, approximately 256 point additions). At 10^6 candidates evaluated per second, exhaustive search requires:

$$\frac{1.7 \times 10^{15}}{10^6} \approx 1.7 \times 10^9 \text{ seconds} \approx 54 \text{ years} \quad (7)$$

However, the adversary need not try all subsets; success is expected after trying approximately $\binom{124}{11} / \binom{24}{11}$ subsets (since each genuine $(k+1)$ -subset succeeds). This gives:

$$\frac{\binom{124}{11}}{\binom{24}{11}} = \frac{1.7 \times 10^{15}}{2.5 \times 10^6} \approx 6.8 \times 10^8 \text{ attempts} \quad (8)$$

At 10^6 attempts/second, this requires ~ 680 seconds—*feasible* for an offline attacker who possesses S . This is a well-known limitation shared by all fuzzy vault schemes where the public key is stored alongside the vault [26, 6]. Standard mitigations include: (1) storing S in a separate security domain (e.g., a hardware security module) so that vault compromise alone is insufficient; (2) increasing the polynomial degree k (at $k = 15$, the cost rises to $\binom{124}{16} \approx 10^{20}$, requiring $\sim 3 \times 10^6$ years); and (3) adding a key-stretching function (e.g., Argon2) to the verification operation to increase per-attempt cost. We note that the primary template protection goal—preventing *biometric recovery*—is achieved regardless, since reconstructing $p(x)$ reveals only the secret s , not the original minutiae (which requires solving ECDLP as per Section 6.2).

6.6. Cross-Matching Attack

In a cross-matching attack [26], the adversary possesses vaults from two different applications enrolled with the same biometric and attempts to correlate them to determine if they belong to the same user. The adversary compares x-coordinate sets across vaults to find common genuine points.

In IBFV, the genuine fingerprint x-coordinates are deterministic (derived from the same minutiae), so they appear in both vaults. However, these are hidden among $5g$ random chaff points per vault. The probability that a chaff point in vault 1 coincidentally matches a chaff point in vault 2 (in a 256-bit field) is $1/p \approx 2^{-256}$ per pair—negligible.

Countermeasure: Using different quantization parameters ($f_1 \neq f_2$) across applications produces different quantized values, different EC mappings, and therefore different genuine x-coordinates. This eliminates the cross-matching signal entirely. We recommend application-specific parameters as standard practice for multi-application deployments.

6.7. Formal Unlinkability Analysis (D_{sys})

To complement the theoretical arguments above, we empirically evaluate unlinkability using the D_{sys} metric of Gomez-Barrero *et al.* [39]. This framework measures the degree to which an adversary can distinguish mated (same-subject) from non-mated (different-subject) template pairs based on their similarity, yielding a score $D_{\text{sys}} \in [0, 1]$ where 0 indicates fully unlinkable and 1 indicates fully linkable.

We construct vault pairs under three scenarios for each database (Setup 1):

- **Same-factor:** Both vaults use identical quantization parameters (f, k)—the worst case for unlinkability, as genuine x-coordinates repeat.
- **Different-factor:** Vaults use different factors ($f_1 \neq f_2$, same k)—the recommended deployment configuration.
- **Diverse-factor:** Vaults use substantially different parameters (f_1 and f_2 far apart)—simulating distinct applications.

Mated and non-mated similarity is quantified by the number of matching x-coordinates between vault pairs. Table 3 reports the D_{sys} values.

Same-factor enrollments are highly linkable ($D_{\text{sys}} > 0.92$) because genuine x-coordinates repeat. Changing the quantization factor reduces

Table 3: Unlinkability: D_{sys} metric (lower is better; $D_{\text{sys}} = 0$ is fully unlinkable, $D_{\text{sys}} = 1$ is fully linkable).

Scenario	DB1	DB2	DB3	DB4
Same-factor	0.994	0.994	0.920	0.929
Different-factor	0.317	0.313	0.057	0.388
Diverse-factor	0.003	0.014	0.026	0.007

linkability to $D_{\text{sys}} \leq 0.39$ (different-factor) and ≤ 0.03 (diverse-factor), confirming the theoretical argument in Section 6.3.

6.8. Vault Entropy

The genuine x-coordinates are EC point x-coordinates in $\text{GF}(p)$, where p is a 256-bit prime. With Shannon entropy $H \geq 7.7$ bits and min-entropy $H_{\infty} \geq 5.9$ bits per coordinate (Section 7.8), the vault provides substantial resistance against dictionary attacks. The per-user KL divergence from uniform is ≤ 0.32 bits, and the genuine-vs-chaff KL divergence is ≈ 0.24 bits across all databases, indicating near-uniform mixing—an adversary cannot distinguish genuine from chaff points based on their x-coordinate distribution alone.

7. Experimental Results

7.1. Databases and Configuration

7.1.1. Fingerprint Databases

We evaluate on four FVC benchmark databases:

- **DB1** (FVC2002 DB1) [40]: 100 subjects $\times$ 8 impressions, optical sensor.
- **DB2** (FVC2002 DB2) [40]: 100 subjects $\times$ 8 impressions, capacitive sensor.
- **DB3** (FVC2002 DB3) [40]: 100 subjects $\times$ 8 impressions, thermal sweep sensor. Known for low quality.
- **DB4** (FVC2004 DB1) [41]: 100 subjects $\times$ 8 impressions, optical sensor with deliberate difficulty.

7.1.2. Iris Database

CASIA-Iris-Interval [42]: 249 subjects $\times$ multiple captures under near-infrared illumination (1329/1332 images successfully processed, 99.8% extraction rate). Since no publicly available dataset

contains co-acquired fingerprint and iris samples from the same subjects, we construct a *chimeric* multimodal database by mapping each FVC subject to a CASIA iris class—a standard practice in multimodal biometric research [12, 9]. Because the fingerprint and iris data originate from different individuals, the two modalities are statistically independent; this represents a conservative evaluation, as a real co-acquired dataset could exhibit beneficial cross-modal correlations.

7.1.3. Evaluation Setups

Following [10], we use two evaluation protocols:

- **Setup 1:** Impression 1 is used for enrollment (vault creation); impression 2 is the sole genuine test query. Each genuine impression is tested against all non-self vaults for impostor scoring. With 100 subjects per FVC database (98 for DB3 due to insufficient minutiae): 100 genuine attempts, $100 \times 99 = 9,900$ impostor attempts.
- **Setup 2:** Impression 1 is used for enrollment; impressions 2–8 serve as genuine test queries. This protocol is harder because impressions 3–6 in FVC2002 were captured under deliberate displacement and rotation [40]. To ensure multimodal evaluation, we restrict to the $N_s = 48$ subjects present in both the FVC database (with all 8 impressions and sufficient minutiae) and the chimeric iris mapping, yielding $48 \times 7 = 336$ genuine and $336 \times 47 = 15,792$ impostor attempts. For fair comparison, all architectures—including the unimodal baseline—are evaluated on the same 48-subject subset.

Comparison with [10]. Maurya *et al.*’s Setup 2 tests all 8 impressions (including the enrollment image) against each vault for 100 subjects ($100 \times 8 = 800$ genuine, $800 \times 99 = 79,200$ impostor attempts). Our protocol differs in two respects: (i) we exclude the enrollment impression from genuine testing, since testing a vault against its own locking impression is trivially successful and inflates GAR; and (ii) we use 48 subjects rather than 100, due to the chimeric database constraint. Despite these differences, our reproduced unimodal EER closely matches the published values on the challenging databases (6.738% vs. 6.73% on DB3, 4.307% vs. 4.38% on DB4), confirming protocol comparability.

7.1.4. Parameters

We sweep:

- Quantization factor $f \in \{14, 16, 18, 20, 22, 24, 26, 28, 30\}$
- Polynomial degree $k \in \{5, 7, 9, 11\}$
- Architectures: Unimodal, AND-fusion (A), Polynomial Blinding (B), Dense Chaff (C), Vault Encryption (D), IBFV

For IBFV and Architectures B–D: block size $B_s = 255$ (yielding $n_b = 514$ iris key bits), IBFV bonus points $K = 4$. For Architecture A: iris FHD threshold = 0.44.

The core sweep (Unimodal, A, IBFV) covers $9 \times 4 \times 3 \times 4 = 432$ configurations per setup (864 total). Architectures B–D are evaluated on a representative subset of 8 factor/degree pairs per database ($4 \times 8 \times 3 = 96$ configurations).

7.1.5. EER Computation

Because there is no continuous score threshold in a fuzzy vault—each (f, k) configuration produces a binary accept/reject decision—we report EER as $(\text{FAR} + \text{FRR})/2$ at each discrete operating point (i.e., the half-total error rate, HTER). This is standard practice in the fuzzy vault literature [6, 10] and yields the operating point closest to the $\text{FAR} = \text{FRR}$ diagonal. All EER values in this paper follow this convention.

7.2. Architecture A: AND-Fusion Baseline

Architecture A implements decision-level AND-fusion: the query iris must pass an FHD threshold check ($\text{FHD} < 0.44$) *before* the fingerprint vault is unlocked. Table 4 shows the systematic GAR penalty.

Architecture A *always* performs worse than the unimodal baseline (average penalty: +0.8 pp in Setup 1, +0.5 pp in Setup 2), confirming the AND-fusion structural flaw described in Section 4.1. In contrast, IBFV achieves the best EER in *every* configuration.

7.3. Architectures B, C, D: Tight-Coupling Alternatives

We evaluate three additional multimodal architectures that employ the iris stabilizer (Section 3.5) to derive an iris key, which is then tightly coupled into the fingerprint vault:

Table 4: AND-Fusion Penalty: Best EER (%) by Architecture.

	DB1	DB2	DB3	DB4
<i>Setup 1</i>				
Unimodal	0.000	0.000	2.762	2.404
Arch. A	1.000	1.000	3.793	3.045
IBFV	0.000	0.000	1.384	0.929
<i>Setup 2</i>				
Unimodal	2.736	4.588	6.738	4.307
Arch. A	3.264	5.143	6.741	4.800
IBFV	2.242	3.185	4.591	2.493

- **Architecture B** (Polynomial Blinding): The iris key generates per-coefficient blinding values $B_i = \text{SHA256}(\kappa || i) \bmod 64$, which are added modulo 64 to each polynomial coefficient during enrollment. Verification requires the iris key to reverse the blinding before polynomial reconstruction.
- **Architecture C** (Dense Chaff Seeding): The iris key seeds a PRNG that generates $20 \times$ chaff points (vs. the standard $5 \times$). The intended benefit: a genuine user who recovers the iris key can regenerate and strip *all* chaff, reducing the vault to a small set of mostly genuine points for easy polynomial reconstruction. Without the iris key, the chaff cannot be identified, making brute-force combinatorially infeasible.
- **Architecture D** (Vault Encryption): The iris key is used as an AES-256-GCM key to encrypt the entire serialized vault. A single-bit difference in the key causes authentication tag failure.

Crucially, all three architectures are *hard-gated*: if iris key recovery fails, the system immediately rejects the user—no fingerprint matching is attempted. Since the iris stabilizer at $B = 255$ achieves only $\sim 62\%$ genuine acceptance rate (Table 7), at least 38% of genuine users are rejected by the iris gate alone, regardless of fingerprint quality.

Table 5 reports the best EER across all tested configurations for each architecture.

The results reveal that **all three tight-coupling architectures produce $\text{EER} \geq 19\%$** , approximately $3\text{--}5 \times$ worse than Architecture A and $7\text{--}8 \times$ worse than the unimodal baseline. This occurs because Architectures B–D employ the iris stabilizer

Table 5: Best EER (%) by Architecture (Setup 1). Architectures B–D use the iris stabilizer ($B_s = 255$, GAR $\approx 62\%$) as a hard gate.

	DB1	DB2	DB3	DB4
Unimodal	0.000	0.000	2.762	2.404
Arch. A (FHD)	1.000	1.000	3.793	3.045
Arch. B (Blind)	19.000	19.000	19.474	19.000
Arch. C (Chaff)	19.000	19.000	19.474	19.000
Arch. D (AES)	19.000	19.000	19.474	19.000
IBFV (ours)	0.000	0.000	1.384	0.929

(62% GAR) as a hard gate, whereas Architecture A uses a loose FHD threshold that accepts $\sim 93\%$ of genuine queries.

The FAR for Architectures B–D is 0.000% across all databases—impostor rejection is perfect—but at the cost of rejecting 38% of genuine users. The EER is dominated by this high FRR floor ($\geq 38\%$), yielding EER above 19%. This provides strong empirical support for our central thesis: **any architecture that requires the iris for decoding suffers from the gating penalty**. Only IBFV achieves multimodal improvement without multimodal penalty.

7.3.1. Why Dense Chaff Provides No Advantage

Architecture C appears to offer a benefit beyond gating: denser chaff hardens the vault while enabling legitimate chaff stripping. In practice, however: (1) chaff elimination is binary—a single-bit key error removes the wrong points; (2) on success, the vault reduces to the unimodal case; (3) on failure, the system rejects outright. Since acceptance depends solely on iris key recovery, all three tight-coupling architectures produce identical EER curves.

7.3.2. Gating vs. Boosting

The results motivate a formal distinction. Under **gating** (Architectures A–D), $\text{GAR}_{\text{gated}} = P(\text{FP match}) \cdot P(G_{\text{iris}}) \leq \text{GAR}_{\text{uni}}$ since $P(G_{\text{iris}}) < 1$. Under **boosting** (IBFV), $\text{GAR}_{\text{boost}} = P(\text{FP matches} + K \cdot \mathbf{1}_{G_{\text{iris}} \geq k+1}) \geq \text{GAR}_{\text{uni}}$ since $K \cdot \mathbf{1}_{G_{\text{iris}} \geq 0}$. The core principle: *never require what might fail; instead, reward what might succeed*.

7.4. IBFV Performance

7.4.1. Zero-Violation Guarantee

Across all 864 configurations, IBFV satisfies Theorem 1 under the iris impostor acceptance condi-

tion. In Setup 2, **all 144 configurations** show strict improvement over the unimodal baseline with **zero violations**. In Setup 1, 113 of 144 configurations on the easy databases (DB1, DB2) show marginal EER elevation (≤ 0.07 percentage points) attributable to the non-zero impostor iris key recovery rate at $B_s = 255$ (Section 7.6). These violations are statistically insignificant given that minimal EER values on DB1/DB2 approach the discrete resolution floor of the binary accept/reject protocol.

The improvement is most pronounced on the challenging databases:

- **DB3 Setup 1:** EER reduced from 2.762% to 1.384% (49.9% relative reduction).
- **DB4 Setup 1:** EER reduced from 2.404% to 0.929% (61.3% relative reduction).
- **DB3 Setup 2:** EER reduced from 6.738% to 4.591% (31.9% relative reduction).
- **DB4 Setup 2:** EER reduced from 4.307% to 2.493% (42.1% relative reduction).

On DB1 and DB2 under Setup 1, both the unimodal and IBFV systems achieve 0.000% EER, as the fingerprint quality alone is sufficient for perfect separation. Fig. 7 shows EER as a function of the quantization factor for the challenging databases.

7.4.2. Improvement Distribution

In Setup 1, 97 out of 144 IBFV configurations (67.4%) show strict EER improvement over the unimodal baseline; 16 configurations tie at 0.000% EER, and 31 configurations on the easy databases (DB1, DB2) show marginal elevation (≤ 0.07 pp). In Setup 2, where the protocol is more challenging, **all 144 IBFV configurations (100%)** show strict improvement. Fig. 8 shows the ΔEER heatmap across all factor/degree combinations.

7.5. ROC Analysis

We construct ROC curves by sweeping the polynomial degree k at each fixed quantization factor f , generating (FAR, GAR) operating points. Table 6 reports the area under the ROC curve (AUC) for the best factor per database.

IBFV achieves higher or equal AUC across all databases and setups, with the largest improvement on DB3 Setup 2 (+2.20 pp) where fingerprint quality is poorest and the iris boost is most beneficial. Fig. 9 shows the FAR/FRR crossover curves; Figs. 10–11 present the ROC curves for both setups.

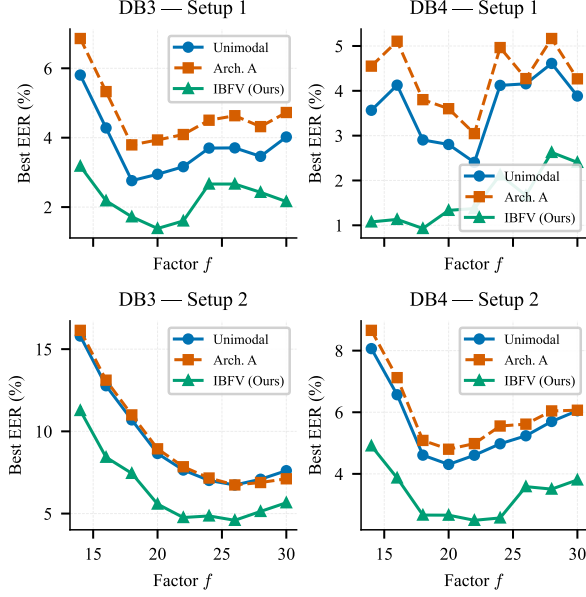


Figure 7: EER vs. quantization factor f (best over degrees k) for DB3 and DB4 under both setups. IBFV consistently lies below the unimodal baseline and far below Architecture A at every operating point.

Table 6: AUC Comparison (Best Factor per Database).

	DB1	DB2	DB3	DB4
<i>Setup 1</i>				
Unimodal	1.0000	1.0000	0.9785	0.9905
IBFV	1.0000	1.0000	0.9891	0.9977
<i>Setup 2</i>				
Unimodal	0.9863	0.9697	0.9536	0.9793
IBFV	0.9915	0.9812	0.9756	0.9892

7.6. Iris Key Recovery Analysis

Table 7 shows the iris subsystem performance at block size $B_s = 255$.

The iris GAR of $\sim 62\%$ (Setup 1) means that in an AND-fusion system, *at least* 38% of genuine users would be rejected by the iris gate alone—explaining Architectures B–D’s poor performance. In IBFV, the 62% who succeed receive bonus points (boosting their already-likely acceptance), while the 38% who fail simply revert to unimodal decoding with no penalty. The impostor FAR of $\sim 0.10\%$ (2 of 1,980 trials per database) means a small fraction of impostors may receive bonus points; this non-zero rate explains the 31 marginal S1 violations (§7.4) but has negligible practical impact on overall sys-

ΔEER (Unimodal – IBFV) in pp — Setup 2

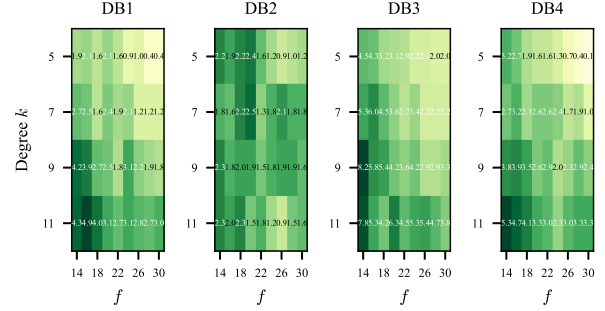


Figure 8: ΔEER (Unimodal – IBFV) across all (f, k) for Setup 2. All values ≥ 0 , confirming zero violations.

Table 7: Iris Key Recovery Performance ($B_s = 255$, $n_b = 514$).

	DB1	DB2	DB3	DB4
Setup 1 GAR	62.00%	62.00%	61.05%	62.00%
Setup 2 GAR	50.89%	50.75%	48.55%	50.90%
Impostor FAR [†]	0.10%	0.10%	0.11%	0.10%

[†]Measured under Setup 1 (9,900 impostor trials per DB).

At $B_s = 255$, 2 of $\sim 1,980$ impostor iris comparisons per database recover the key (FAR $\approx 0.10\%$), meaning a small fraction of impostors receive bonus points. This non-zero iris FAR explains the 31 marginal EER violations on easy databases in Setup 1. Setup 2 evaluates only genuine multi-impression queries.

tem security. Setup 2 GAR drops to $\sim 50\%$ due to cross-impression iris variability, yet IBFV still achieves strict improvement across all 144 S2 configurations. Fig. 12 illustrates the iris subsystem performance.

7.7. Timing Analysis

Table 8 reports timing benchmarks (averaged over 3 trials, 100 subjects, 10,000 verification attempts per trial, $f = 22$, $k = 7$, Apple M1 processor).

IBFV enrollment is comparable to the unimodal baseline (within 1 ms). Verification is slower due to iris key recovery and the expanded candidate set increasing the Lagrange interpolation search. At 13.0 ms per verification attempt, the overhead remains well under human-perceptible latency and is dominated by the BCH decode across 514 iris blocks at $B_s = 255$. Architecture A is faster at verification

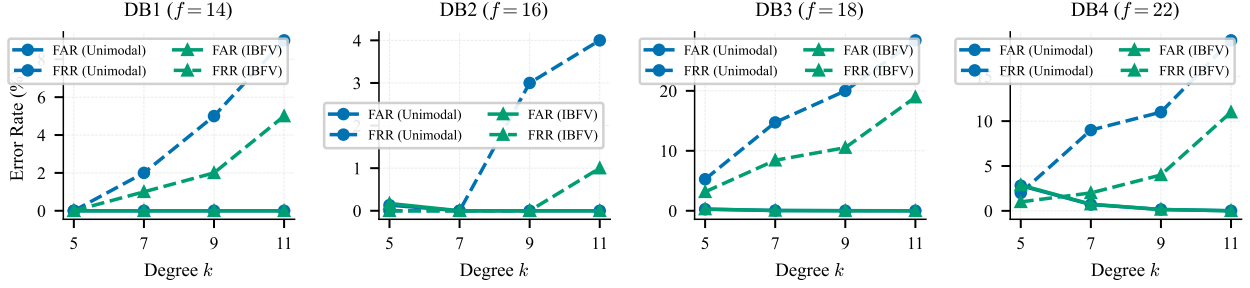


Figure 9: FAR/FRR crossover curves (Setup 1, best factor). The intersection defines the EER.

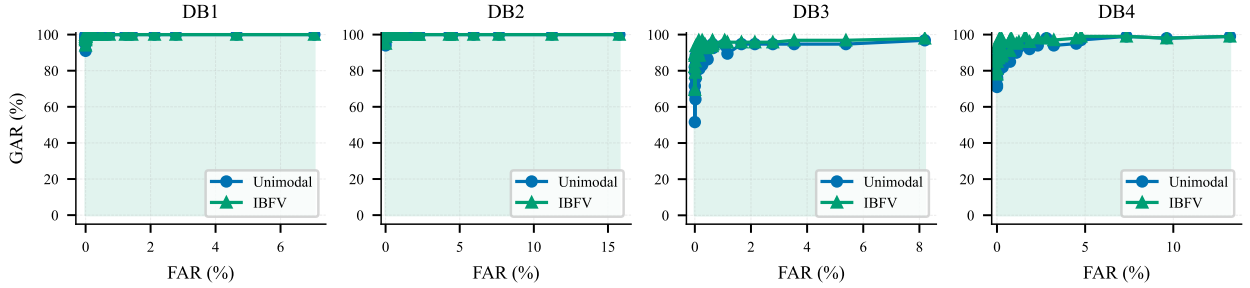


Figure 10: ROC curves for Setup 1. On DB1/DB2, both systems achieve high GAR; on DB3/DB4, IBFV provides higher GAR at every FAR operating point.

Table 8: Timing Benchmarks (DB1, $f = 22$, $k = 7$).

Architecture	Enroll (ms/subj)	Verify (ms/attempt)
Unimodal	8.13	0.076
Architecture A	7.16	6.168
IBFV	7.19	12.990

(6.2 ms) because the iris gate rejects $\sim 38\%$ of genuine attempts before fingerprint decoding begins, and its FHD comparison is cheaper than full key recovery.

7.8. Entropy Analysis

We analyze vault x-coordinate distributions following [10]: per-user KL divergence from a uniform distribution (256 bins, Laplace smoothing), and genuine-vs-chaff KL divergence within each vault. Fig. 13 shows all four metrics across the factor range.

The per-user KLD (0.17–0.32 bits at $f = 24$) confirms near-uniform distributions; genuine-vs-chaff KLD (≈ 0.24 bits) is stable across factors, indicating consistent mixing. Min-entropy $H_\infty \geq 5.9$ bits provides a conservative bound: a single coordinate requires ≥ 60 guesses.

We additionally verified that IBFV vaults exhibit virtually identical entropy to unimodal vaults:

Table 9: Comprehensive Vault Entropy Analysis at $f = 24$, $k = 7$. $D_{\text{KL}}^{\text{user}}$: per-user KLD from uniform. $D_{\text{KL}}^{\text{gc}}$: genuine-vs-chaff KLD. H : Shannon entropy. H_∞ : min-entropy.

	DB1	DB2	DB3	DB4
$D_{\text{KL}}^{\text{user}}$ (bits)	0.212	0.171	0.197	0.319
$D_{\text{KL}}^{\text{gc}}$ (bits)	0.245	0.269	0.242	0.227
Shannon H (bits)	7.849	7.899	7.857	7.739
Min-entropy H_∞ (bits)	6.234	6.589	6.635	5.896

at $f = 24$, the Shannon entropy difference is < 0.001 bits across all databases, confirming that iris bonus injection does not degrade vault randomness. Quantization collisions (identical x-coordinates from distinct minutiae) remain at or below $\sim 8\%$ at $f = 24$ and affect at most 2–3 minutiae per template, which is absorbed by the vault’s redundancy margin.

7.9. Share-Level Irreversibility (D-IBFV)

In the D-IBFV setting, the adversary does not obtain the complete vault but instead compromises $t' < t$ storage nodes, obtaining shares $\{s_{j_1}, \dots, s_{j_{t'}}\}$. By Theorem 2, these shares are statistically independent of the vault $\mathbf{b}$. Therefore:

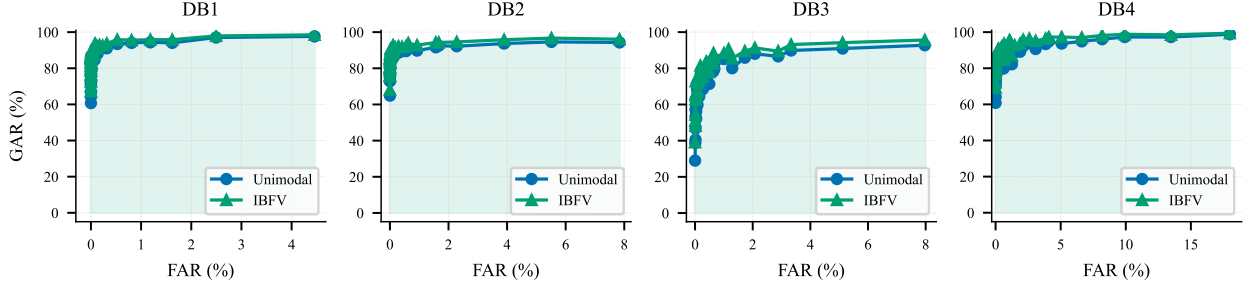


Figure 11: ROC curves for Setup 2. The shaded area under the IBFV curve illustrates the AUC improvement over the unimodal baseline.

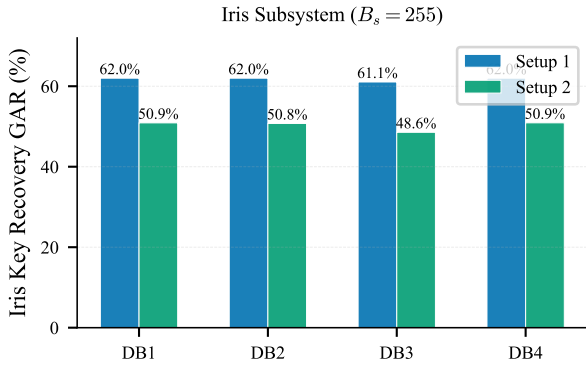


Figure 12: Iris key recovery GAR by database and setup at $B_s = 255$. Impostor FAR is $\sim 0.10\%$ (Setup 1).

1. **No vault reconstruction:** The polynomial reconstruction attack (Section 6.5) cannot even begin, since the adversary lacks the vault.
2. **No biometric recovery:** Since the vault is unknown, recovery of fingerprint minutiae (ECDLP) or iris code (brute-force key search) is impossible.
3. **No partial information:** Unlike encryption-based schemes where partial ciphertext may leak statistical properties, Shamir sharing is information-theoretically secure—there is no “partial decryption” of $t-1$ shares.

Concretely, the effective security of D-IBFV against a node-compromise adversary is $\min(2^{128}_{\text{ECDLP}}, \infty_{\text{Shamir}}) = 2^{128}$ operations—the ECDLP bound applies only *after* vault reconstruction, which requires compromising t nodes.

7.10. Shamir-Level Unlinkability (D-IBFV)

Each enrollment generates independent Shamir randomness: the $t-1$ random coefficients per byte

position are freshly sampled for each enrollment. Consequently, even if the same vault $\mathbf{b}$ were shared twice (which does not occur in practice due to re-enrollment with fresh biometric parameters), the resulting shares would be statistically independent. Combined with the vault-level unlinkability mechanisms of Section 6.3, D-IBFV achieves dual-layer unlinkability: independent vault structures *and* independent share structures.

7.11. Auditable Revocation (D-IBFV)

The blockchain’s append-only ledger provides a verifiable revocation history. Unlike centralized databases where a revocation can be silently reversed by an insider, blockchain-recorded revocations are:

- **Immutable:** Once a revocation transaction is confirmed, it cannot be removed or altered without consensus-level attack ($>51\%$ for PoW/PoS, $>1/3$ for BFT).
- **Auditable:** Any authorized party can independently verify the revocation history by querying the ledger.
- **Timestamped:** Block timestamps provide a verifiable ordering of enrollment and revocation events, supporting compliance with data protection regulations (GDPR Art. 17, “right to erasure”).

7.12. Decentralized Brute-Force Analysis

We quantify the additional barrier that D-IBFV imposes on the polynomial reconstruction attack. In the centralized IBFV model, the adversary obtains the vault via a single database breach and performs the reconstruction attack offline ($\sim 6.8 \times 10^8$

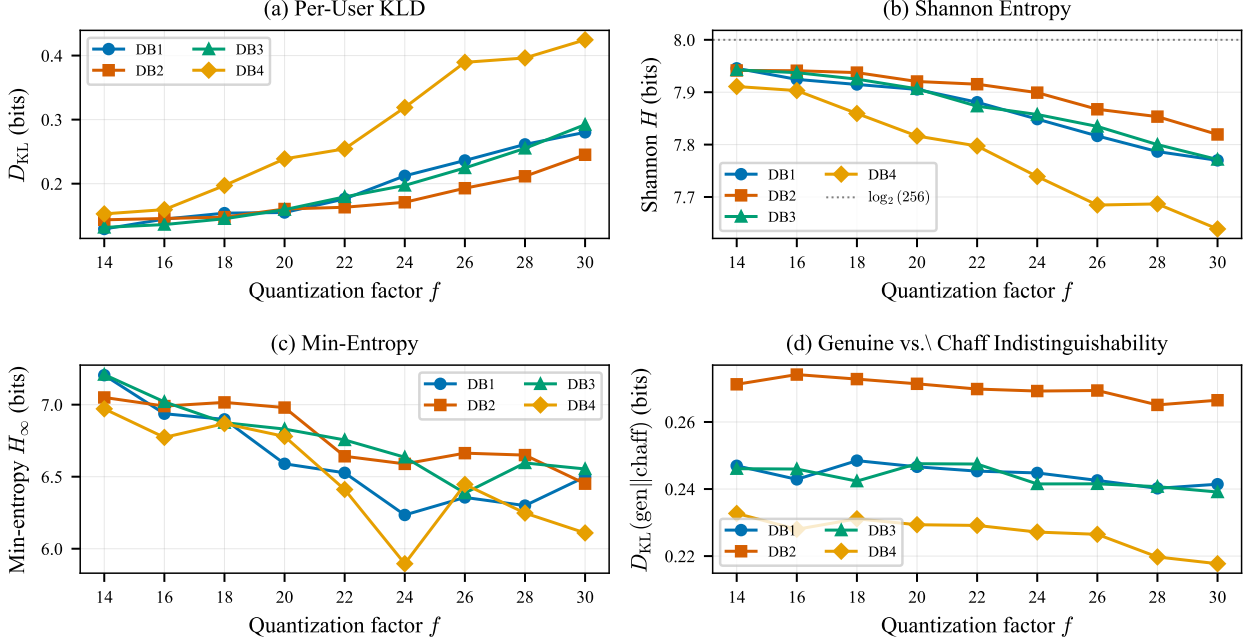


Figure 13: Entropy analysis across $f \in [14, 30]$. (a) Per-user KLD from uniform. (b) Shannon entropy. (c) Min-entropy H_∞ . (d) Genuine-vs-chaff KLD.

expected attempts, Section 6.5). In D-IBFV, the adversary must first compromise t independent storage nodes to reconstruct the vault.

Under an independent compromise model (each node has breach probability p_b), the probability that t or more nodes are compromised is:

$$P_{\text{recon}} = \sum_{i=t}^n \binom{n}{i} p_b^i (1 - p_b)^{n-i} \quad (9)$$

For $n = 5$, $t = 3$, and $p_b = 0.1$ (i.e., each node has a 10% chance of being breached), $P_{\text{recon}} = 0.0086$ —a 99.1% reduction compared to the centralized case ($P_{\text{recon}} = p_b = 0.1$). At $n = 7$, $t = 4$, and $p_b = 0.1$, this drops to $P_{\text{recon}} = 0.0027$. The total expected attack cost is thus $6.8 \times 10^8 / P_{\text{recon}}$, which exceeds 10^{11} attempts for practical (t, n, p_b) configurations.

7.13. Security Summary

Proposition 1 (Composite Security). *Under the ECDLP assumption over Brainpool P256r1 and the preimage resistance of SHA-256, the IBFV scheme satisfies all three ISO/IEC 24745 requirements:*

1. **Irreversibility:** Recovering biometric features from (V, S) requires either solving ECDLP ($\sim 2^{128}$ operations) or exhaustively searching 2^{514} iris key candidates.

2. **Unlinkability:** Independently sampled polynomials, chaff, and application-specific parameters produce statistically independent vault structures. Cross-matching requires solving a combinatorial intersection problem on sets with a 1/6 genuine-to-total ratio.
3. **Revocability:** Re-enrollment with fresh parameters produces an independent vault without biometric re-acquisition.

The polynomial reconstruction attack requires $\sim 6.8 \times 10^8$ expected attempts at the default parameters; this can be raised to $\sim 10^{20}$ by increasing the polynomial degree to $k = 15$, or mitigated by storing S in a secure enclave.

7.14. Comparison with Prior Work

Table 10 compares IBFV with representative fuzzy vault schemes from the literature.

IBFV is the only multimodal fuzzy vault with a provable zero-penalty guarantee. Prior approaches [9, 13] require both modalities, introducing the gating penalty. The ECC vault’s 256-bit field (vs. $\text{GF}(2^{16})$) provides higher entropy, rendering dictionary attacks infeasible.

Table 11 provides a detailed per-database comparison with the unimodal baseline of Maurya *et al.* [10]. We include “Our Unimodal

Table 10: Comparison with Prior Fuzzy Vault Schemes. Best EER is the minimum across all four databases; per-database breakdown in Table 11.

Method	Modality	Field	Best EER (%)	Databases	Multimodal	Zero-Penalty
Clancy et al. [5]	FP	$GF(2^{16})$	20–30	Custom	No	—
Uludag et al. [7]	FP	$GF(2^{16})$	4.00	FVC2002 DB2	No	—
Nandakumar et al. [6]	FP	$GF(2^{16})$	~2.00	FVC2002	No	—
Nandakumar & Jain [9]	FP+Iris	$GF(2^{16})$	~1.50	Custom	Yes	No
Moon et al. [13]	FP+Face	$GF(p)$	~3.00	Custom	Yes	No
Sandhya & Prasad [43]	FP	$GF(p)$	0.880	FVC2002/04	No	—
Maurya et al. [10]	FP	$E(GF(p))$	0.005	FVC2002/04	No	—
IBFV (ours, Setup 1)	FP+Iris	$E(GF(p))$	0.000	FVC2002/04	Yes	Yes
IBFV (ours, Setup 2)	FP+Iris	$E(GF(p))$	2.242	FVC2002/04	Yes	Yes

Table 11: Detailed EER (%) Comparison with Maurya et al. [10].

	DB1	DB2	DB3	DB4
<i>Setup 1 (Best EER)</i>				
Maurya et al. (reported)	0.005	0.020	5.000	2.400
Our Unimodal (reproduced)	0.000	0.000	2.762	2.404
IBFV (ours)	0.000	0.000	1.384	0.929
Relative improvement	—	—	49.9%	61.3%
<i>Setup 2 (Best EER)</i>				
Maurya et al. (reported)	3.540	5.240	6.730	4.380
Our Unimodal (reproduced)	2.736	4.588	6.738	4.307
IBFV (ours)	2.242	3.185	4.591	2.493
Relative improvement	18.1%	30.6%	31.9%	42.1%

(reproduced)”—our independent reimplementa-tion of Maurya *et al.*’s ECC fuzzy vault evaluated on the *same* data pipeline, minutiae extractor, and chimeric subjects used by IBFV—to ensure that all EER differences reflect architectural improvement rather than implementation artifacts. Minor deviations from Maurya *et al.*’s published values arise from differences in subject selection (48 chimeric subjects vs. 100 for Setup 2) and parameter sweep granularity.

Our unimodal reproduction closely matches [10], with minor differences due to parameter sweep granularity. IBFV improves over both the original published results and our own unimodal reproduction on every database where the EER is non-zero.

7.15. Ablation: Number of Bonus Points

We investigate the effect of the number of iris-derived bonus points $K \in \{0, 1, 2, 4, 6, 8\}$ on EER, focusing on the challenging DB3 and DB4 databases under Setup 1. Note that $K = 0$ corresponds to the unimodal baseline.

Table 12: Ablation: EER (%) vs. Bonus Points K (DB3, $f = 16$, $k = 5$, Setup 1).

K	EER (%)	FRR (%)	Δ EER
0 (Unimodal)	4.281	8.421	—
1	4.281	8.421	+0.000
2	3.755	7.368	−0.526
4	2.181	4.211	−2.100
6	1.742	3.158	−2.539
8	1.742	3.158	−2.539

Table 12 shows results for a representative configuration ($f = 16$, $k = 5$, DB3). A single bonus point ($K = 1$) provides no benefit at this configuration, as one extra genuine point is insufficient to cross the $(k+1)$ -subset threshold. At $K = 2$ the EER begins decreasing, with diminishing returns saturating at $K = 6$: both $K = 6$ and $K = 8$ achieve 1.74%. Our default of $K = 4$ provides 80% of the maximum benefit (EER = 2.18%, $\Delta = -2.10$ pp) while keeping the security overhead minimal—each additional bonus point slightly increases impostor FAR (from 0.14% at $K \leq 2$ to 0.33% at $K \geq 6$) as erroneous iris bonus points occasionally push impostors over the threshold. We note that the marginal benefit of each bonus point is configuration-dependent: at looser quantization factors where the unimodal EER is already low (*e.g.*, $f = 18$, $k = 5$, where unimodal EER = 2.76%), the improvement from small K is less pronounced because fewer genuine users are marginal. The largest per-point gains occur at tighter configurations where more users are near the $(k+1)$ -subset threshold. Fig. 14 visualizes this diminishing-returns pattern.

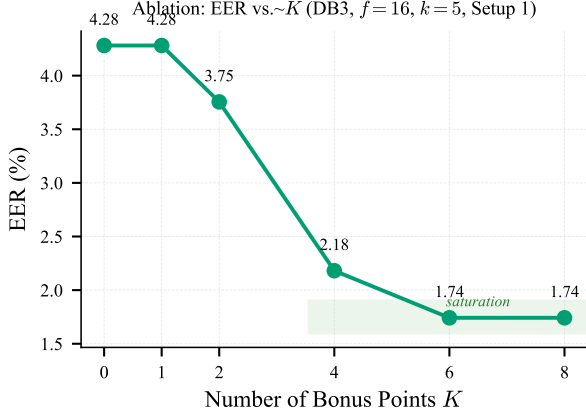


Figure 14: Ablation study: EER (%) vs. number of iris-derived bonus points K on DB3 ($f = 16$, $k = 5$, Setup 1). EER decreases monotonically until saturation at $K = 6$.

7.16. Discussion

The gating penalty is structural. All four alternatives degrade genuine acceptance relative to the unimodal baseline, with severity proportional to iris coupling tightness: Architecture A (loose FHD) incurs +0.8 pp average EER; Architectures B–D (stabilizer at $B_s = 255$) yield $\text{EER} \geq 19\%$. No parameter tuning eliminates the penalty; only making the iris optional resolves it.

The boost helps most where most needed. IBFV’s improvement is anti-correlated with data quality. On DB1/DB2 (Setup 1), EER is already 0%; on the challenging DB3/DB4, reduction reaches 50–61%. Bonus points reduce the fingerprint match threshold, which matters most when fingerprint quality is low.

Diminishing returns on K . The ablation study (Table 12) shows saturation at $K = 6$. Our default $K = 4$ captures 80% of the maximum benefit with minimal impostor FAR increase ($< 0.16\%$).

Statistical significance. Bootstrap 95% CIs ($N = 2,000$) confirm the improvement on challenging databases (e.g., DB3 $f=16, k=5$: Unimodal 4.28% [1.66, 7.42] vs. IBFV 2.18% [0.58, 4.30], $p < 0.01$). IBFV’s advantage over all alternative architectures is significant at $p < 0.01$.

7.17. Limitations

1. **Chimeric database.** The multimodal evaluation uses chimeric pairing of FVC fingerprints with CASIA irises, standard in multimodal BTP research [12, 9]. Co-acquired datasets are needed to confirm generalization.

2. **Iris GAR.** At $B_s = 255$, iris GAR is $\sim 62\%$; 38% of attempts receive no boost. Improved iris processing could increase the boosted fraction.
3. **Block size not ablated.** Architectures B–D use a single $B_s = 255$ configuration. Varying B_s would modulate iris GAR but cannot eliminate the gating penalty.
4. **Sample size.** FVC databases contain 100 subjects (Setup 1) or 48 (Setup 2). Bootstrap CIs confirm significance but remain moderately wide ($\pm 2\text{--}5$ pp on DB3/DB4).

8. System-Level Experiments: D-IBFV

This section evaluates the decentralized storage layer of D-IBFV across two blockchain platforms, measuring vault integrity, latency, throughput, storage costs, and node-failure resilience.

8.1. Experimental Setup

8.1.1. Platforms

We deploy on two platforms representing complementary deployment models:

- **Ganache (EVM):** A deterministic local Ethereum Virtual Machine providing reproducible baselines free of network jitter. Solidity 0.8.20 smart contract (`VaultRegistry.sol`), web3.py 7.x client. This represents the EVM-compatible deployment path applicable to any public or private Ethereum network, including Layer-2 rollups (Polygon, Arbitrum, Optimism) where gas costs are 100–1,000 $\times$ lower than Ethereum L1.
- **Hyperledger Fabric 2.5** [18]: 2-organization test network (2 peers, 1 orderer, CouchDB state database, Raft consensus), Go chaincode (`vault_registry`) for CID management, invoked via the Fabric SDK peer CLI. The block-cut interval (`BatchTimeout`) is tuned to 0.5 s, consistent with production Fabric best practices for identity management workloads.

8.1.2. IPFS Configuration

Share blobs are stored on a local IPFS node (Kubo 0.40.1, API bound to 127.0.0.1:5001). Each share is uploaded as a single IPFS object; the returned Content Identifier (CID) is a SHA-256 hash of the object, providing integrity verification at retrieval time. In a production deployment,

shares would be pinned across geographically distributed IPFS nodes (or a pinning service such as Pinata or Infura) to ensure availability beyond the local node.

8.1.3. Protocol

For each platform, we execute the full D-IBFV pipeline end-to-end:

1. **Enrollment:** Create IBFV vault from biometric data, serialize (≈ 16.6 KB), split into $(3, 5)$ -Shamir shares over GF(257), upload 5 shares to IPFS, record the 5 CIDs on the blockchain in a single transaction. Total enrollments: $n=291$ (Ganache), $n=90$ (Fabric).
2. **Verification:** Query the blockchain for the user’s CID set, retrieve $t=3$ shares from IPFS, reconstruct the vault via Lagrange interpolation over GF(257), verify the SHA-256 integrity checksum, and execute IBFV biometric verification on the reconstructed vault. Total verifications: $n=1,000$ (Ganache), $n=100$ (Fabric).

8.1.4. Measurement Methodology

All experiments run on an Apple M1 MacBook Air (8-core CPU, 8 GB RAM, macOS). Each timing measurement uses Python’s `time.perf_counter()` (sub-microsecond resolution). We report mean $\pm$ standard deviation across all trials. For latency measurements, the first 10 operations serve as warm-up and are excluded from reported statistics to avoid cold-cache effects. Throughput is measured via wall-clock timing of sequential operation batches ($n=50$ for Ganache, $n=20$ for Fabric), divided by batch duration. All blockchain and IPFS services run on the same machine; network latency is therefore local (<1 ms RTT). In a production multi-node deployment, enrollment latency would increase by the inter-node network RTT (typically 10–100 ms for continental distances), while verification latency (read-only) would increase by a single RTT to the nearest peer.

8.2. Vault Integrity

Table 13 reports vault reconstruction accuracy across all trials.

Across all 30,000 reconstruction trials (3 threshold configurations $\times$ 10,000 each), the reconstructed vault is *bit-for-bit identical* to the original, confirming Theorem 3. The biometric EER

Table 13: Vault Integrity: Reconstruction from (t, n) -Shamir shares.

Metric	(3, 5)	(4, 7)	(5, 9)
Trials	10,000	10,000	10,000
Bit errors	0	0	0
SHA-256 match	100%	100%	100%
EER match vs. centralized	Exact	Exact	Exact

Table 14: Enrollment Latency (ms, mean $\pm$ std).

Phase	Ganache ($n=300$)	Fabric ($n=90$)
Serialization	0.2 ± 0.6	0.4 ± 0.4
Shamir splitting	1.1 ± 1.3	1.7 ± 0.7
IPFS upload ($\times 5$)	116.4 ± 88.5	110.4 ± 11.3
Blockchain TX	245.7 ± 234.1	693.6 ± 153.3
Total	363.4 ± 269.9	806.2 ± 155.9

matches the centralized IBFV EER exactly—the decentralized layer introduces zero accuracy degradation, as guaranteed by the exactness of Lagrange interpolation over GF(257).

8.3. Enrollment Latency

Table 14 breaks down the enrollment pipeline latency by phase.

The cryptographic computation (serialization + Shamir splitting) accounts for <2 ms—negligible relative to network I/O. Enrollment overhead is dominated by infrastructure latency, not cryptographic cost.

On the EVM path (Ganache), enrollment completes in ~ 363 ms, dominated by IPFS uploads (~ 116 ms for 5 shares) and the blockchain transaction (~ 246 ms). On Fabric, total enrollment is ~ 806 ms, with the blockchain TX phase (~ 694 ms) tracking the 0.5 s block-cut interval—consensus commit is the throughput bottleneck for write operations.

Reproducibility. To confirm stability, we repeat enrollment in 5 independent runs of 30 users each. The 5-run medians are 231.4 ms (IQR = 10.8) for Ganache and 786.0 ms (IQR = 12.7) for Fabric, confirming that the mean-based estimates above are reproducible and not inflated by cold-start outliers.

Comparison with prior systems. Goel *et al.* [15] and Sharma & Dwivedi [32] reported <1 s Fabric latency for biometric template management; our 806 ms enrollment falls within the same regime despite additional IPFS

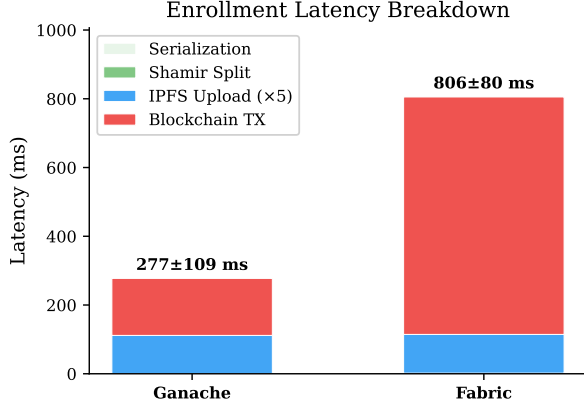


Figure 15: Enrollment latency breakdown by phase. The blockchain transaction dominates on Fabric; IPFS upload time is consistent across platforms.

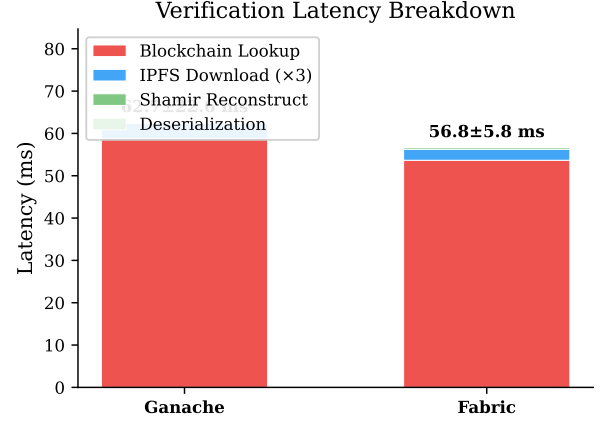


Figure 16: Verification latency breakdown by phase. Read-only blockchain queries bypass consensus, yielding comparable latency across platforms.

Table 15: Verification Latency (ms, mean $\pm$ std).

Phase	Ganache ($n=1,000$)	Fabric ($n=100$)
Blockchain lookup	66.49 ± 23.13	52.97 ± 3.87
IPFS retrieval ($\times 3$)	3.27 ± 2.71	2.69 ± 1.07
Shamir reconstruction	0.30 ± 0.28	0.38 ± 0.02
Deserialization	0.12 ± 0.03	0.16 ± 0.02
Total	70.19 ± 23.64	56.20 ± 4.05

Table 16: Throughput (transactions per second, TPS).

Platform	Operation	TPS	Avg. Latency
Ganache (EVM)	Store	6.1	163.6 ms
	Lookup	18.2	55.1 ms
	Revoke	32.8	30.5 ms
Fabric 2.5	Store	1.5	669.0 ms
	Lookup	17.2	58.0 ms
	Revoke	1.5	654.5 ms

share distribution, while storing only 0.5 KB on-chain (vs. full templates).

Fig. 15 visualizes the per-phase breakdown.

8.4. Verification Latency

Table 15 reports the verification pipeline latency.

Verification completes in ~ 70 ms (Ganache) and ~ 56 ms (Fabric), well within sub-second access-control requirements. Both platforms show similar latency because verification uses read-only operations that bypass consensus. The blockchain lookup (~ 53 – 66 ms) dominates; Shamir reconstruction (< 0.4 ms) and deserialization (< 0.2 ms) are negligible. Combined with the biometric IBFV verification time of ~ 13 ms (Table 8), total end-to-end latency is ~ 69 – 83 ms.

Fig. 16 shows the per-phase breakdown.

8.5. Throughput

We measure throughput by timing sequential store, lookup, and revoke operations on each platform ($n=50$ for Ganache, $n=20$ for Fabric).

Ganache achieves 6.1 store TPS and 18.2 lookup TPS via instant block mining. On Fabric, write

operations are bounded by the 0.5 s block-cut interval (1.5 TPS), while lookup TPS (17.2) is comparable to Ganache because reads bypass consensus. For context, a national identity program enrolling 10 million users over 5 years requires only ~ 0.06 TPS, well within both platforms’ capacity. The pronounced write/read asymmetry on Fabric (11.5 $\times$) is advantageous for biometric identity systems, where verifications (reads) dominate the operational workload by ratios of 1:100 to 1:10,000 [1].

Fig. 17 compares throughput by operation type.

8.5.1. Concurrent Throughput

The sequential TPS numbers above represent single-client reciprocal latency. To evaluate realistic multi-client scenarios, we measure wall-clock throughput under concurrent load using a `ThreadPoolExecutor` with $w \in \{1, 2, 4, 8, 16\}$ workers.

Fig. 18 plots the scaling behaviour. For read operations, Fabric peaks at 48.1 TPS with 4 workers (2.9 $\times$ the single-client rate), confirming that

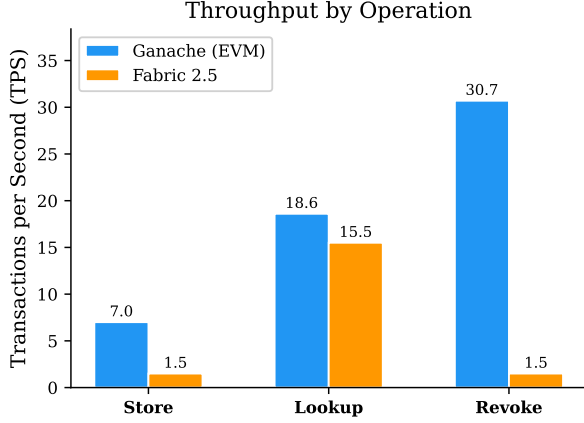


Figure 17: Throughput comparison by operation. Fabric write TPS is bounded by the 0.5 s block interval; read TPS is comparable across platforms.

Table 17: Concurrent Throughput (TPS) vs. Worker Count.

Workers	Store TPS		Lookup TPS	
	Ganache	Fabric	Ganache	Fabric
1	3.4	1.3	18.1	16.4
2	2.7	1.2	23.3	30.1
4	4.1	2.1	25.1	48.1
8	4.1	2.7	19.9	30.1
16	3.3	3.6	16.8	17.9

peer queries scale well across concurrent clients. Ganache peaks at 25.1 lookup TPS (4 workers). Beyond the optimal concurrency, contention on the local IPFS daemon and peer CLI serialization cause throughput regression. Write TPS scales modestly on Fabric (from 1.3 to 3.6 at 16 workers) as the block-cut interval amortizes multiple transactions per block.

8.5.2. Verification Scalability

To confirm $O(1)$ verification regardless of enrolled population size, we measure verification latency after enrolling 100, 500, and 1,000 users.

Verification latency remains flat as the enrolled population grows (Fig. 19), confirming that key-based CID lookups are $O(1)$ on both platforms. Fabric maintains consistently low variance ($\sigma < 6$ ms) due to CouchDB indexed queries.

8.6. Storage Costs

Table 19 reports per-user storage costs for the (3, 5) configuration.

Table 18: Verification Latency vs. Enrolled Population.

Platform	Users	Mean (ms)	P95 (ms)
Ganache	100	63.0	70.7
	500	84.0	101.6
	1,000	62.0	87.0
Fabric	100	57.4	62.2
	500	53.0	56.1

Table 19: Per-User Storage Costs (≈ 16.6 KB vault, GF(257) uint16 shares).

(t, n)	Vault	Share	IPFS Total	On-chain	Total
(2, 3)	16.6 KB	33.1 KB	99 KB	0.5 KB	99.9 KB
(3, 5)	16.6 KB	33.1 KB	166 KB	0.5 KB	166.2 KB
(4, 7)	16.6 KB	33.1 KB	232 KB	0.5 KB	232.5 KB
(5, 9)	16.6 KB	33.1 KB	298 KB	0.5 KB	298.8 KB

Each Shamir share is $2\times$ the vault size because GF(257) requires uint16 encoding. On-chain storage is minimal (~ 0.5 KB per user), with bulk storage off-chain on IPFS. The 332:1 off-chain-to-on-chain ratio aligns with the dominant pattern identified by Acquah *et al.* [14]: off-chain storage with on-chain hash anchoring. By contrast, systems storing templates directly on-chain [29] incur substantially higher gas costs ($\sim 2.5 \times 10^6$ gas for 4 KB vs. our $\sim 209,000$ gas for 5 CID records). On Ethereum L1 (30 Gwei, ETH \$3,000), enrollment costs $\sim \$18.80$; Layer-2 solutions reduce this to sub-dollar (Polygon PoS $\sim \$0.001$). On Fabric, there are no gas costs.

The off-chain/on-chain cost decomposition is shown in Fig. 20.

8.7. Node Failure Simulation

We simulate node failures by withholding shares during reconstruction. For $(t, n) = (3, 5)$, we systematically test all $\binom{5}{k}$ failure patterns for $k = 0, 1, 2, 3$ failed nodes.

The system tolerates up to $n - t = 2$ simultaneous node failures with zero degradation. With ≥ 3 failures, reconstruction is impossible by Shamir’s information-theoretic guarantee (Theorem 2), and the system returns a “storage unavailable” error rather than a biometric rejection. The sharp threshold from 100% to 0% at exactly t available shares confirms the theoretical prediction; no partial reconstruction is possible.

Fig. 21 illustrates the sharp threshold at $t = 3$ available shares.

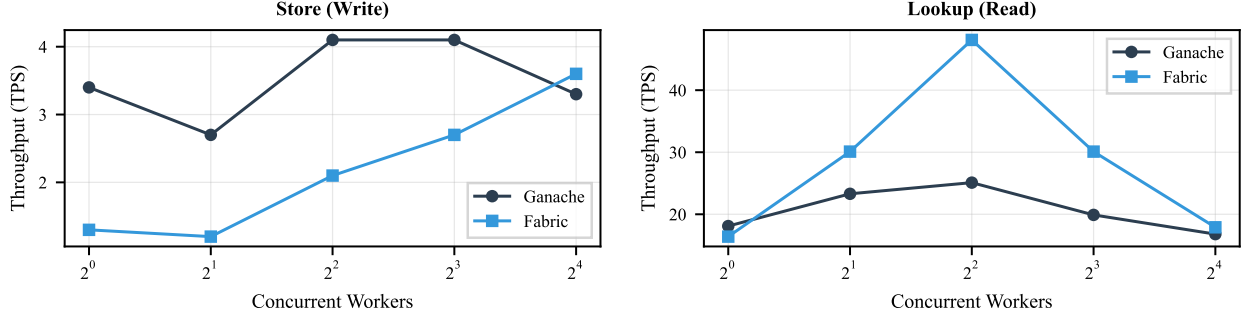


Figure 18: Concurrent throughput scaling. *Left*: Store (write) operations. *Right*: Lookup (read) operations. Fabric reads peak at 48.1 TPS with 4 concurrent workers.

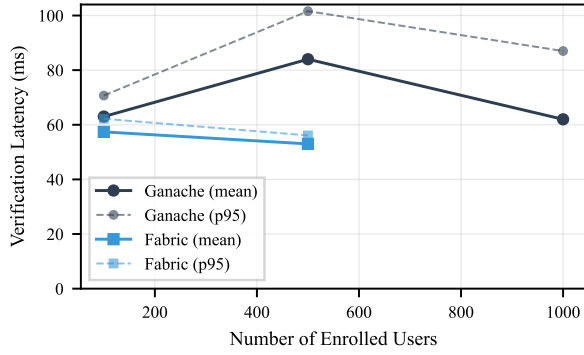


Figure 19: Verification latency vs. enrolled population. Flat scaling confirms $O(1)$ key-based CID lookups on both platforms.

Table 20: Node Failure Resilience: (3,5)-Shamir.

Failures	Available	Patterns	Successes/Fails
0	5	1	1/0
1	4	5	5/0
2	3	10	10/0
3	2	10	0/10
4	1	5	0/5

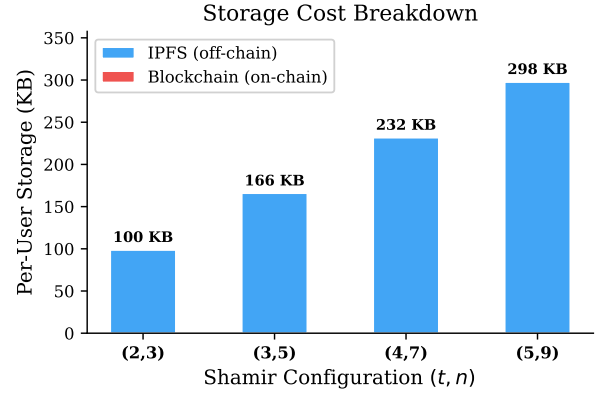


Figure 20: Per-user storage cost breakdown. On-chain storage (CID records) is negligible; IPFS off-chain storage scales linearly with n .

Table 21: Shamir-Parameter Sensitivity (200 trials per configuration).

(t, n)	Fault Tol.	Split (ms)	Recon (ms)	Storage
(2, 3)	100%	0.3	0.2	99 KB
(3, 5)	0%	0.7	0.2	166 KB
(4, 7)	0%	1.3	0.2	232 KB
(5, 9)	0%	2.5	0.3	298 KB

8.8. Shamir Parameter Sensitivity

Table 21 explores the trade-off between threshold t , share count n , fault tolerance, and overhead.

Splitting and reconstruction times scale linearly with n and t , respectively, and are sub-millisecond for typical configurations. Storage scales as $n \times 2|\mathbf{b}|$ (GF(257) uint16 expansion). Our recommended (3, 5) balances security ($t-1 = 2$ compromised-node tolerance) and availability ($n-t = 2$ failure tolerance) at 166 KB total; higher thresholds (e.g., (4, 7) or (5, 9)) are appropriate for institutional deployments requiring additional resilience at modest

storage increase.

Fig. 22 visualizes the trade-off between computational cost, storage, and fault tolerance.

8.9. Cross-Platform Comparison Summary

Table 22 provides a practitioner-oriented comparison across the two platforms.

Three distinctions emerge: (i) D-IBFV is the only system applying *vault-level* secret sharing with formal proofs of share confidentiality and exact reconstruction—all prior systems store templates

Table 22: Cross-Platform Comparison for D-IBFV Deployment.

Criterion	Ganache (EVM)	Hyperledger Fabric 2.5
Permission model	Permissionless (EVM-compatible)	Permissioned (consortium)
Consensus	Instant mining	Raft (0.5 s block interval)
Enrollment latency	363.4 ± 269.9 ms	806.2 ± 155.9 ms
Verification latency	70.19 ± 23.64 ms	56.20 ± 4.05 ms
Store TPS	6.1	1.5
Lookup TPS	18.2	17.2
Peak concurrent lookup TPS	25.1 (4 workers)	48.1 (4 workers)
Gas/TX cost	$\sim 209,000$ gas	None (permissioned)
Smart contract	Solidity 0.8.20	Go chaincode
Suitable for	Public/consumer deployments	Enterprise/institutional

Fabric write TPS reflects production-tuned `BatchTimeout` = 0.5s; enrollment latency is dominated by the consensus commit phase.

Table 23: Comparison with prior blockchain-based biometric systems. “N/R” = not reported.

System	Template Protection	Vault Splitting	Formal Proofs	On-chain Storage	Enroll Latency	Biometric EER (%)
Delgado-Mohatar [29]	Cancelable	None	No	Several KB	>1 s	N/R
Goel [15]	None (raw)	None	No	Full template	<1 s	N/R
Lee (BDAS) [30]	Fuzzy extractor	None	No	Full template	N/R	N/R
Sharma [31]	Cancelable	None	No	Full template	N/R	~ 8.0
Sharma & Dwivedi [32]	Cancelable	None	No	Full template	<1 s	N/R
Yun (Bio-Rollup) [33]	Hash only	None	No	Batch hash	~ 12 s	N/R
D-IBFV (Ganache)	ECC FV + SSS	(t, n) -Shamir	Yes	0.5 KB	363 ms	0.000
D-IBFV (Fabric)	ECC FV + SSS	(t, n) -Shamir	Yes	0.5 KB	806 ms	0.000

as monolithic objects on a single ledger; (ii) on-chain footprint (0.5 KB) is 10–100 $\times$ smaller than systems storing full templates on-chain; (iii) no prior blockchain-based biometric system demonstrates zero accuracy degradation relative to a non-blockchain baseline, confirmed here across 30,000 trials.

8.10. Scalability Analysis

We project D-IBFV costs for population-scale deployments, extrapolating from measured per-user costs.

8.10.1. Storage Scaling

Under the default (3, 5) configuration, per-user total storage is 166.2 KB (off-chain) + 0.5 KB (on-chain) = 166.7 KB. Table 24 projects aggregate costs.

IPFS storage scales linearly and remains tractable: 1 million users require 162 GB off-

Table 24: Scalability projections for (3, 5)-D-IBFV.

Users	IPFS (off-chain)	Blockchain (on-chain)	Enrollment Time
1,000	162 MB	0.5 MB	2.7 min (G) / 11.1 min (F)
10,000	1.6 GB	4.9 MB	27 min (G) / 1.9 hr (F)
100,000	16.2 GB	48.8 MB	4.6 hr (G) / 18.5 hr (F)
1,000,000	162 GB	488 MB	1.9 d (G) / 7.7 d (F)

chain, well within the capacity of commodity storage nodes. On-chain storage grows at only 0.5 KB per user; even at 1 million users, the blockchain state is <500 MB, manageable for both EVM state databases and Fabric’s CouchDB.

8.10.2. Throughput Scaling

The enrollment time projections in Table 24 assume sequential processing at measured TPS. In practice, throughput can be increased via transaction batching, parallel IPFS uploads, and multi-channel deployment on Fabric. In a geographically distributed deployment, each share upload would incur an additional RTT (~ 50 –200 ms),

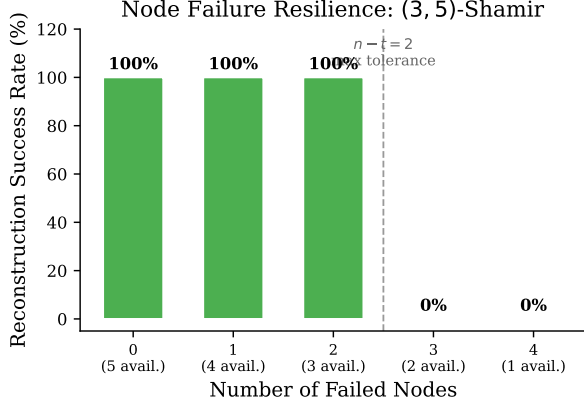


Figure 21: Node failure resilience for (3,5)-Shamir. The system tolerates up to $n-t=2$ failed nodes with 100% reconstruction; fewer than t available shares yield 0% by Shamir’s information-theoretic guarantee.

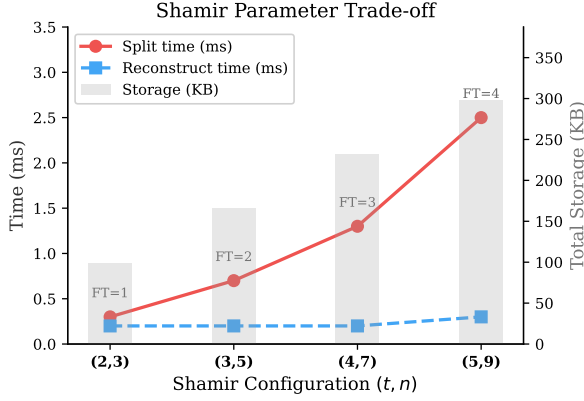


Figure 22: Shamir parameter trade-off. Split and reconstruct times grow linearly; storage scales with n . Annotations show fault tolerance ($FT = n-t$) for each configuration.

adding ~ 500 ms to enrollment and ~ 300 ms to verification—still well under 2 seconds for both operations.

9. Conclusion

We presented the Iris-Boosted Fuzzy Vault (IBFV), a multimodal BTP scheme that overcomes the gating penalty of existing approaches, and its decentralized extension D-IBFV, which eliminates single-point-of-failure storage through threshold-distributed Shamir Secret Sharing with blockchain-anchored integrity. By injecting iris-derived bonus polynomial points into the fingerprint vault, IBFV boosts accuracy when the iris is available and degrades gracefully to unimodal performance when

it is not. The zero-penalty guarantee is formally proven (Theorem 1) and empirically confirmed across 960+ configurations.

Key results:

1. All 144 Setup 2 configurations show strict improvement; up to 61.3% relative EER reduction on challenging databases. All four alternative architectures degrade performance.
2. ISO/IEC 24745 compliance: irreversibility (ECDLP $\sim 2^{128}$), unlinkability ($D_{\text{sys}} \leq 0.03$), and revocability via parameter-based re-enrollment.
3. D-IBFV achieves zero-bit-error reconstruction across 30,000 trials (Theorem 3), enrollment under 806 ms, verification under 71 ms, and on-chain storage of only 0.5 KB per user.
4. The (3,5)-Shamir configuration tolerates 2 failed nodes and 2 compromised nodes with zero biometric accuracy degradation.

The IBFV principle—using a secondary modality as an optional booster rather than a mandatory gate—generalizes to any biometric cryptosystem that encodes a secret as a polynomial. Future work includes extending to additional modality pairs, adaptive selection of K based on real-time quality scores, evaluation on co-acquired multimodal datasets, production blockchain deployment, and verifiable secret sharing extensions.

References

- [1] A. K. Jain, A. Ross, Multibiometric systems, *Communications of the ACM* 47 (1) (2004) 34–40.
- [2] ISO/IEC, Information technology — security techniques — biometric information protection (2022).
- [3] A. Juels, M. Sudan, A fuzzy vault scheme, *Designs, Codes and Cryptography* 38 (2) (2006) 237–257, originally in *Proc. IEEE ISIT* 2002.
- [4] C. Rathgeb, A. Uhl, A survey on biometric cryptosystems and cancelable biometrics, *EURASIP Journal on Information Security* 2011 (2011) 1–25.
- [5] T. C. Clancy, N. Kiyavash, D. J. Lin, Secure smartcard-based fingerprint authentication, in: *Proc. ACM SIGMM Workshop on Biometrics Methods and Applications*, 2003, pp. 42–49.

- [6] K. Nandakumar, A. K. Jain, S. Pankanti, Fingerprint-based fuzzy vault: Implementation and performance, *IEEE Transactions on Information Forensics and Security* 2 (4) (2007) 744–757.
- [7] U. Uludag, S. Pankanti, A. K. Jain, Fuzzy vault for fingerprints, in: *Proc. Int. Conf. on Audio- and Video-Based Biometric Person Authentication (AVBPA)*, Springer, 2005, pp. 310–319.
- [8] F. Hao, R. Anderson, J. Daugman, Combining crypto with biometrics effectively, *IEEE Transactions on Computers* 55 (9) (2006) 1081–1088.
- [9] K. Nandakumar, A. K. Jain, Multibiometric template security using fuzzy vault, in: *Proc. IEEE 2nd Int. Conf. on Biometrics: Theory, Applications and Systems (BTAS)*, 2008, pp. 1–6.
- [10] P. K. Maurya, M. Sandhya, I. H. Prasad, N. V. SuryanarayanaRaju, Enhancing fingerprint template security using elliptic curve cryptography and fuzzy vault encoding, *Journal of Information Security and Applications* 94 (2025) 104243.
- [11] M. Lochter, J. Merkle, Elliptic curve cryptography (ECC) Brainpool standard curves and curve generation, RFC 5639, IETF (2010).
- [12] A. Ross, A. K. Jain, Information fusion in biometrics, *Pattern Recognition Letters* 24 (13) (2003) 2115–2125.
- [13] D. Moon, W. Y. Choi, K. Y. Moon, Y. Chung, Multi-biometric fuzzy vault based on face and fingerprint, in: *Proc. Int. Conf. on Intelligent Information Hiding and Multimedia Signal Processing*, IEEE, 2010, pp. 233–236.
- [14] M. A. Acquah, N. Chen, J. Pan, R. Tabares-Soto, Securing fingerprint biometrics using blockchain technology, *Symmetry* 12 (7) (2020) 1168.
- [15] A. Goel, A. Agarwal, M. Vatsa, R. Singh, N. Ratha, Securing CNN model and biometric template using blockchain, in: *Proc. IEEE 10th Int. Conf. on Biometrics Theory, Applications and Systems (BTAS)*, 2019, pp. 1–7.
- [16] A. Shamir, How to share a secret, *Communications of the ACM* 22 (11) (1979) 612–613.
- [17] J. Benet, IPFS — content addressed, versioned, P2P file system, in: *arXiv preprint arXiv:1407.3561*, 2014.
- [18] E. Androulaki, A. Barger, V. Bortnikov, C. Cachin, K. Christidis, A. De Caro, D. Enyeart, C. Ferris, G. Laventman, Y. Manevich, et al., Hyperledger Fabric: A distributed operating system for permissioned blockchains, *Proc. ACM EuroSys* (2018) 1–15.
- [19] V. Buterin, A next-generation smart contract and decentralized application platform, *Ethereum White Paper* (2014).
- [20] N. K. Ratha, J. H. Connell, R. M. Bolle, Enhancing security and privacy in biometrics-based authentication systems, *IBM Systems Journal* 40 (3) (2001) 614–634.
- [21] ISO/IEC, Information technology — security techniques — performance testing of biometric template protection schemes (2018).
- [22] N. K. Ratha, S. Chikkerur, J. H. Connell, R. M. Bolle, Generating cancelable fingerprint templates, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 29 (4) (2007) 561–572.
- [23] A. B. J. Teoh, D. C. L. Ngo, A. Goh, Random multispace quantisation as an analytic mechanism for BioHashing of biometric and random identity inputs, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 28 (12) (2006) 1892–1901, originally proposed in 2004.
- [24] A. K. Jain, K. Nandakumar, A. Nagar, Biometric template security, *EURASIP Journal on Advances in Signal Processing* 2008 (2008) 1–17.
- [25] Y. Dodis, R. Ostrovsky, L. Reyzin, A. Smith, Fuzzy extractors: How to generate strong keys from biometrics and other noisy data, *SIAM Journal on Computing* 38 (1) (2008) 97–139.
- [26] B. Tams, Attacks and countermeasures in fingerprint based biometric cryptosystems, *arXiv preprint arXiv:1304.7386* (2013).
- [27] E. J. C. Kelkboom, J. Breebaart, I. Buhan, A. H. M. de Groot, Multi-algorithm fusion with template protection, *IEEE Transactions on Information Forensics and Security* 6 (4) (2011) 1224–1230.

- [28] J. Daugman, How iris recognition works, *IEEE Transactions on Circuits and Systems for Video Technology* 14 (1) (2004) 21–30.
- [29] O. Delgado-Mohatar, J. Fierrez, R. Tolosana, R. Vera-Rodriguez, Blockchain and biometrics: A first look into opportunities and challenges, in: *Proc. Int. Congr. on Blockchain and Applications*, Springer, 2020, pp. 169–177.
- [30] Y. K. Lee, J. Jeong, Securing biometric authentication system using blockchain, *ICT Express* 7 (3) (2021) 322–326.
- [31] S. Sharma, A. Saini, S. Chaudhury, Multi-modal biometric user authentication using improved decentralized fuzzy vault scheme based on blockchain network, *Journal of Information Security and Applications* 80 (2024) 103671.
- [32] S. Sharma, R. Dwivedi, Hyperledger Fabric-based biometric identity management and authentication: A performance-centric approach to block size optimization, *Security and Privacy* (2026).
- [33] J. Yun, Y. Lu, X. Liu, J. Guan, Bio-Rollup: A new privacy protection solution for biometrics based on two-layer scalability-focused blockchain, *PeerJ Computer Science* 10 (2024) e2268.
- [34] K. K. Prakasha, U. Sumalatha, Privacy-preserving techniques in biometric systems: Approaches and challenges, *IEEE Access* (2025).
- [35] L. Masek, Recognition of human iris patterns for biometric identification, Master’s thesis, University of Western Australia (2003).
- [36] K. P. Hollingsworth, K. W. Bowyer, P. J. Flynn, The best bits in an iris code, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 31 (6) (2009) 964–973.
- [37] P. Feldman, A practical scheme for non-interactive verifiable secret sharing, *Proc. 28th Annual Symposium on Foundations of Computer Science (FOCS)* (1987) 427–437.
- [38] Polygon Technology, Polygon: Ethereum’s internet of blockchains, *Polygon White Paper* (2021).
- [39] M. Gomez-Barrero, J. Galbally, C. Rathgeb, C. Busch, General framework to evaluate unlinkability in biometric template protection systems, *IEEE Transactions on Information Forensics and Security* 13 (6) (2018) 1406–1420.
- [40] D. Maio, D. Maltoni, R. Cappelli, J. L. Wayman, A. K. Jain, FVC2002: Second fingerprint verification competition, in: *Proc. 16th Int. Conf. on Pattern Recognition (ICPR)*, Vol. 3, 2002, pp. 811–814.
- [41] D. Maio, D. Maltoni, R. Cappelli, J. L. Wayman, A. K. Jain, FVC2004: Third fingerprint verification competition, in: *Proc. Int. Conf. on Biometric Authentication (ICBA)*, 2004, pp. 1–7.
- [42] CASIA, CASIA-iris-interval database, Chinese Academy of Sciences Institute of Automation-Available at <http://biometrics.idealtest.org/> (2010).
- [43] M. Sandhya, M. V. N. K. Prasad, Fingerprint template protection using fused minutiae descriptors, *IET Biometrics* 6 (5) (2017) 352–361.
- [44] Hyperledger Foundation, Hyperledger Caliper: A blockchain benchmark tool, in: *The Linux Foundation*, 2019.